IVIVC App User Guide

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1 Introduction

Establishing an in vitro-in vivo correlation (IVIVC) can support the evaluation of medical devices that include absorbable materials by relating in vivo changes in material properties over time to corresponding in vitro test results, such as degradation profiles. Potential uses of IVIVCs include evaluating manufacturing or material changes and supporting predictions of in vivo performance when modifying an existing device design. This tool is intended to facilitate IVIVC development. The user supplies paired in vitro and in vivo fitting data; an optional in vitro prediction dataset can also be supplied. The app applies selected preprocessing steps, fits and compares multiple correlation approaches and model families, evaluates selected performance metrics and cross-validation results, generates fitting and prediction plots, and produces report-ready summaries for user review and external documentation.

The app fits empirical relationships between the in vitro and in vivo time courses supplied by the user. The scope includes absorbable polymers that degrade by hydrolysis or enzymatic action, as well as absorbable metals. It does not directly model material chemistry, device geometry, implantation conditions, or degradation mechanisms.

This guide describes the app and offers considerations for reviewing its outputs. The app provides information to support user review of models; it does not select or recommend a model or determine regulatory acceptability. The app and user guide are advisory and do not establish requirements, acceptance criteria, or a prescribed model-selection process. Users are responsible for determining how to document their model-selection rationale and other analysis decisions, consistent with the intended use and applicable credibility considerations.

For general background on credibility assessment and reporting of computational modeling studies in medical device submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

For additional discussion of model performance, cross-validation results, uncertainty, residual diagnostics, and model selection considerations, see Appendix A. For an example of interpreting model performance, see Appendix B. For definitions of commonly used terms, see Appendix D.

2 What the app does

- Applies selected normalization and scaling, and uses interpolation where required for paired-data approaches

and mappings.

- Fits user-selected IVIVC approaches and model families to paired in vitro and in vivo data.
- Calculates selected goodness-of-fit, model-comparison, and cross-validation metrics.
- Generates fitting, comparison, and prediction plots with uncertainty displays where supported.
- Creates summaries for user-selected approaches or models for downstream review and documentation.

3 Getting started

The following provides a brief orientation for a first run of the app. The settings described here as defaults are software defaults only; they do not establish preferred analysis choices or regulatory expectations. Detailed descriptions and interpretation considerations are provided in the sections that follow.

Minimum input. A fitting dataset is required whose first four columns contain in vitro time, in vitro response, in vivo time, and in vivo response. A separate in vitro prediction dataset, whose first two columns contain time and response, is optional. Additional columns can contain measurement uncertainty and replicate counts.

From setup to results. On the setup page, upload the fitting dataset and, if desired, a prediction dataset. Select at least one analysis approach and, for model-based approaches, at least one candidate model. Review or modify the preprocessing, model-fitting, performance-metric, cross-validation, and diagnostic-plot options as applicable, then select **Analyze**. The app processes the uploaded data, performs the selected analyses, and opens the results page, where results are organized by processing, fitting, model comparison, and prediction outputs as applicable.

Optional measurement uncertainty. To use supplied SD or SEM, select Approach 2 on its own and expand **Measurement uncertainty**. See Section 4.2 for the input choices and assumptions.

Current software defaults. No analysis approaches or candidate models are selected initially. Measurement uncertainty starts as **Not supplied** for all datasets. No normalization or scaling is selected. Shared-grid interpolation is selected for approaches that use matched quantities. The default displayed metrics are R^2, RMSE, and AIC_c. Shuffle-split cross-validation is selected by default. The available CV-fit-variability, raw-residual, and Q–Q diagnostic plots are not enabled by default. These defaults provide a starting configuration and can be changed to match the dataset and analysis purpose.

Main outputs. After analysis, the results include the processed data used by each approach, fitted or mapped relationships, selected performance and cross-validation summaries, diagnostic plots, model-comparison information where applicable, and prediction outputs when a prediction dataset is supplied. User-selected results can also be assembled into a downloadable summary report.

4 Inputs

The IVIVC app requires paired in vitro and in vivo degradation data, along with user-selected analysis settings that define how the data are processed, modeled, and summarized. At a high level, the inputs include:

- a spreadsheet containing in vitro and in vivo time-course degradation data for fitting correlations;
- optionally, a spreadsheet containing in vitro data from which to generate predictions using fitted correlations;
- selection of one or more IVIVC modeling approaches;
- selection of candidate models to evaluate, where applicable;
- optionally, measurement uncertainty for Approach 2;
- optionally, preprocessing options, such as normalization, interpolation, and scaling;
- selected performance metrics to display in the results and reports;
- optionally, cross-validation settings for applicable approaches;
- grid-search settings for model initialization, for which the app provides default values; and

- optionally, advanced analysis plotting settings;

The sections below provide additional detail on input file structure, required data columns, preprocessing options, model selection, and optional analysis settings.

4.1 Input datasets

The app uses separate fitting and prediction datasets. The fitting dataset is required because it provides the paired in vitro and in vivo data used to establish the IVIVC relationship. The prediction dataset is optional and is used only when applying a fitted relationship to new in vitro observations in order to predict the corresponding in vivo behavior.

In this guide, **response value** means the measured y-axis value that changes with time, such as mass remaining, molecular weight, a mechanical property (e.g., tensile or burst strength), or another quantitative degradation-related endpoint. In the input files, these are the in vitro and in vivo value columns.

When interpreting an IVIVC, consider measurement reliability, sample size, comparability of the in vitro and in vivo measurements, and experimental limitations such as incomplete material recovery or contamination, because a good statistical fit does not rule out systematic measurement error.

Input	Description
Fitting dataset	Required dataset used to establish the in vitro/in vivo relationship. The first four columns are interpreted as in vitro time, in vitro response value, in vivo time, and in vivo response value.
Prediction dataset	Optional in vitro dataset used after fitting to generate in vivo predictions. The first two columns are interpreted as in vitro time and in vitro response value.

Supported formats are CSV and Excel (.xls, .xlsx). Include a header row; the required columns are read by position in the order above for either format. For Excel workbooks, select the sheet containing the relevant data. The two calibration profiles need not have matching times or equal lengths; leave unused cells blank when one profile ends earlier.

Additional columns may contain SD/SEM or replicate counts. Give these columns unambiguous headers and enter the relevant header in the uncertainty settings. Keep each error and count on the same row as its response value, including when profiles have different lengths.

4.2 Measurement uncertainty (optional; Approach 2)

The dataset settings describe uncertainty in the entered response values; time coordinates are treated as fixed. They are available only when Approach 2 is selected on its own. In the **Measurement uncertainty** panel, choose an input mode separately for the in vitro calibration, in vivo calibration, and optional in vitro prediction dataset. Any of these datasets can be left without supplied uncertainty.

Supplied calibration errors affect fitting and fitted uncertainty; smaller errors give an observation more influence. Prediction-input errors affect prediction intervals, as described in Section 5.2.

Uncertainty input	What to enter
Not supplied	No error values. A calibration fit estimates its error scale from scatter around the fitted curve; prediction responses without supplied errors are treated as fixed.
Constant SD or SEM	One positive error value that applies to every response used in the analysis of that dataset.
SD or SEM column	The header of a column containing one positive error value for each response used in the analysis.

Choose **Individual observations** when supplying a standard deviation (SD), which describes variation among individual measurements at a given time. Choose **Reported means** when supplying a standard error of the mean

(SEM), which describes uncertainty in a reported mean. The column name does not determine which interpretation the app uses.

If means are accompanied by replicate SD and count n, convert SD to SEM before entry: SEM = SD/$\sqrt{n}$ for independent replicates. The app does not calculate this conversion. A replicate-count column is optional for reported means; its positive integer counts are recorded as metadata. They are not used to divide SEM again or to increase the number of fitted observations.

Enter absolute SD/SEM in the same units as the original response values, before any preprocessing selected in the app. Convert relative errors, such as a percentage of the response, to absolute values first. Every observation used in the analysis needs a finite, strictly positive error when uncertainty is supplied for that dataset. Missing or invalid errors on those rows cause an input error; they do not cause those observations to be silently removed. Error cells may be blank on unused padding rows.

Confirm that errors in the original measurements are independent across entered rows, including measurements at the same time, and that prediction measurements are independent of calibration measurements. Prediction-input uncertainty also requires a prediction dataset. See Section 4.4 for error rescaling and supported preprocessing choices.

Future in vivo measurements. With a prediction dataset, optionally select future measurement variability separately: none (the default), a constant SD/SEM, the prediction inputs' uncertainty, or the in vivo calibration uncertainty at the nearest predicted in vivo time. Reuse requires the corresponding source uncertainty and inherits its individual/mean interpretation. For a constant, choose individual future measurements (SD) or future means (SEM), and enter the error in the original in vivo calibration response units. The app applies the fixed in vivo normalization references once. Future errors are assumed independent of the entered measurements. These options change prediction-response intervals, not fitting; their interpretation is described in Section 5.2.

4.3 Analysis approaches and models

In this app, an analysis **approach** is the overarching method used to establish or evaluate an in vitro/in vivo correlation. A **model** is the mathematical form used within an approach when an equation is needed to describe the relationship.

4.3.1 Approaches

Approaches differ in how the in vitro and in vivo time courses are related to one another. Approach numbers identify the methods and do not indicate a ranking or preference.

Approach	Description	May be relevant when	Available models
Approach 1: Direct Correlation	Approach 1 directly relates the in vitro and in vivo times associated with the same response values after preprocessing (i.e., in vivo time is modeled as a function of in vitro time). When the datasets do not contain the same response values, the app can interpolate time at shared response values to create the paired times used for fitting.	This approach may be relevant when a key analytical question concerns how in vitro time at a given response corresponds to in vivo time at that response.	Double Exponential, Exponential, Linear, Linear (zero intercept), Power

Approach	Description	May be relevant when	Available models
Approach 2: Independent Model Fitting	Approach 2 fits the same model family (e.g., exponential decay) separately to the preprocessed observed in vitro and in vivo datasets. The two datasets do not need matching time points or response values, so interpolation is not used for Approach 2 fitting.	This approach may be relevant when the two datasets were measured on different time schedules, when each dataset is expected to follow the same type of degradation curve, or when a specific model form is scientifically meaningful for both datasets.	Double Exponential, Double Exponential (Normalized), Exponential, Exponential (Normalized), Exponential + Logistic, Exponential + Logistic (Normalized), Finite-Time Power Decay, Finite-Time Power Decay (Normalized), Linear + Power Decay, Linear + Power Decay (Normalized), Logistic, Logistic (Normalized), Logistic + Logistic, Logistic + Logistic (Normalized), Modified Gompertz, Modified Gompertz (Normalized), Stretched Exponential, Stretched Exponential (Normalized)
Approach 3: Direct Mapping	Approach 3 uses direct interpolation-based mappings between datasets rather than fitting a parametric model. Response values interpolated at shared times are used for response-value ratios over time, while times interpolated at shared response values are used for time ratios across response-value ranges.	This approach may be relevant for exploratory comparisons, for datasets where a simple equation does not describe the observed pattern well, or when the user wants a direct data-based mapping rather than a fitted model.	No parametric model selection

When a prediction dataset is supplied, each approach uses the prediction data differently. See Section 5.2 for the available prediction methods and outputs.

For Approach 2 models where a finite value can be calculated, the app summarizes each fitted decay curve with an **integral time constant**. For a fitted model $f(t)$, the time constant is defined as

$$\tau = \frac{1}{f(0)} \int_0^\infty f(t)\, dt,$$

which is the area under the fitted curve after normalizing its fitted value at time zero to 1 [3]. It provides a single characteristic time scale for comparing fitted in vitro and in vivo profiles; a larger value generally corresponds to slower overall decay. For a simple exponential decay, the integral equation yields the time constant. The fitted time constants are also used by one of the optional Approach 2 prediction methods described in Section 5.2.

The time constant is evaluated at the fitted parameters and is reported only when its integral can be calculated. A valid estimate can remain available without an uncertainty interval. See Section 5.3 for how to read the uncertainty displays.

4.3.2 Models

Candidate models are the mathematical equations available for fitting within the applicable approaches. Different model forms represent different assumptions about the shape of the degradation relationship, such as linear behavior, power-law behavior, or sigmoidal transition behavior.

Model	Used in	Functional form	Notes
Double Exponential	Approach 1, Approach 2	$y = ae^{bx} + ce^{dx}$	Sum of two exponential terms. It can represent profiles with two apparent rates, but the added flexibility may require more data to estimate stably.
Double Exponential (Normalized)	Approach 2	$y = we^{bx} + (1 - w)e^{dx}$	Two exponential components with coefficients w and 1-w, fixing the response at time zero to 1. Examples include fraction remaining or retained properties with two apparent rates. The coefficient and both rates remain fitted; selecting this model does not normalize data.
Exponential	Approach 1, Approach 2	$y = ae^{bx}$	Two-parameter exponential relationship. It can describe monotonic change that accelerates or decelerates exponentially over the fitted range.
Exponential (Normalized)	Approach 2	$y = e^{bx}$	Exponential profile with amplitude fixed at 1 for responses already on a relative scale, such as fraction remaining or retained mechanical properties. Only the rate parameter is fitted; selecting this model does not normalize data.
Exponential + Logistic	Approach 2	$y = y_0 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x - t_{50})/s})$	Two-phase response profile combining an early exponential change and a delayed logistic transition, with fitted baseline and magnitudes. Examples include multistage changes in mass remaining, molecular weight, or retained mechanical properties.
Exponential + Logistic (Normalized)	Approach 2	$y = 1 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x - t_{50})/s})$	Exponential-plus-logistic response profile with baseline fixed at 1. Both change magnitudes remain fitted, so the terminal response is not fixed. Examples include fraction remaining or normalized retained properties with early and delayed changes. Selecting this model does not normalize data.

Model	Used in	Functional form	Notes
Finite-Time Power Decay	Approach 2	$y = A + (y_0 - A)\max(1 - kx, 0)^n$	Response profile that reaches zero in finite time, with fitted amplitude, rate, and shape exponent. This surface-erosion form can also be used empirically for remaining mass, volume, molecular weight, or retained mechanical properties when the shape is appropriate.
Finite-Time Power Decay (Normalized)	Approach 2	$y = \max(1 - kx, 0)^n$	Finite-time power-decay profile with amplitude fixed at 1. This surface-erosion form can describe fraction remaining or normalized retained properties when the shape is appropriate. The rate and shape exponent remain fitted; selecting this model does not normalize data.
Linear	Approach 1	$y = a + bx$	Straight-line empirical relationship with an intercept and slope. It is appropriate when the in vitro/in vivo relationship is approximately linear over the fitted range.
Linear (zero intercept)	Approach 1	$y = ax$	Straight-line empirical relationship constrained to pass through zero. It may be useful when a zero in vitro value should correspond to a zero in vivo value under the selected preprocessing and response scale.
Linear + Power Decay	Approach 2	$y = y_0 \max\{1 - kx - (x/t_f)^n, 0\}$	Empirical remaining-response model combining linear and power-law loss, with the initial response fitted and negative remaining values set to zero. For n > 1, the power term can describe accelerating loss. Requires nonnegative time; k >= 0, tf > 0, n > 0, and y0 >= 0.
Linear + Power Decay (Normalized)	Approach 2	$y = \max\{1 - kx - (x/t_f)^n, 0\}$	Linear-plus-power remaining-response model for data already on a 0-1 scale. The initial response is fixed at 1, leaving the linear rate, power-term time scale, and exponent to be fitted. Requires nonnegative time; k >= 0, tf > 0, and n > 0.

Model	Used in	Functional form	Notes
Logistic	Approach 2	$y = y_0 - A/(1 + e^{-(x-t_{50})/s})$	Scaled and shifted logistic (sigmoid) response profile with fitted baseline and change magnitude. Examples include delayed changes in mass remaining, molecular weight, or retained mechanical properties.
Logistic (Normalized)	Approach 2	$y = 1 - 1/(1 + e^{-(x-t_{50})/s})$	Logistic (sigmoid) response profile with baseline and total change fixed at 1, giving asymptotes at 0 and 1. Examples include fraction remaining or normalized retained properties with a delayed transition. The analyzed response must already match this scale; selecting the model does not normalize data.
Logistic + Logistic	Approach 2	$y = A\left[w\frac{L(x;k_1,t_1)}{L(0;k_1,t_1)} + (1-w)\frac{L(x;k_2,t_2)}{L(0;k_2,t_2)}\right], L(x;k,t) = \frac{1}{1+e^{k(x-t)}}$	Weighted sum of two decreasing logistic transitions, with the initial response A fitted and the terminal response fixed at zero. The weight w is the first component's share of the initial response; t1 and t2 locate the transitions, and positive k1 and k2 control their steepness. Requires A >= 0 and 0 <= w <= 1. Examples include two-stage changes in mass remaining or retained mechanical properties. The components are empirical and need not represent physical phases.

Model	Used in	Functional form	Notes
Logistic + Logistic (Normalized)	Approach 2	$y = w\frac{L(x;k_1,t_1)}{L(0;k_1,t_1)} + (1 - w)\frac{L(x;k_2,t_2)}{L(0;k_2,t_2)}$, $L(x;k,t) = \frac{1}{1+e^{k(x-t)}}$	Weighted sum of two decreasing logistic transitions with the initial response fixed at 1 and the terminal response fixed at zero. The weight w is the first component's initial contribution; t1 and t2 locate the transitions, and positive k1 and k2 control their steepness. Requires 0 <= w <= 1. Examples include fraction remaining or normalized retained properties with two transitions. The components are empirical and need not represent physical phases. Selecting this model does not normalize the input data.
Modified Gompertz	Approach 2	$y = y_0 - A\exp\{-\exp[(er_{max}/A)(t_{lag}-x)+1]\}$	Modified Gompertz response profile with an asymmetric sigmoidal transition, fitted baseline, change magnitude, rate, and lag. Examples include mass remaining, molecular weight, or retained properties with a delayed asymmetric change.
Modified Gompertz (Normalized)	Approach 2	$y = 1 - \exp\{-\exp[er_{max}(t_{lag} - x) + 1]\}$	Modified Gompertz response profile with baseline and total change fixed at 1, giving asymptotes at 0 and 1. Examples include fraction remaining or normalized retained properties with a delayed asymmetric transition. Selecting this model does not normalize data.
Power	Approach 1	$y = ax^b$	Power-law empirical relationship. It can describe nonlinear monotonic relationships where the response changes as a fitted power of time or the mapped predictor.
Stretched Exponential	Approach 2	$y = ae^{bx^c}$	Exponential relationship with a fitted stretching exponent. It can describe non-single-rate behavior, but the additional shape parameter can increase sensitivity when data are sparse.

Model	Used in	Functional form	Notes
Stretched Exponential (Normalized)	Approach 2	$y = e^{bx^c}$	Stretched exponential profile with amplitude fixed at 1 for responses already on a relative scale, such as fraction remaining or normalized molecular weight. The rate and stretching exponent remain fitted; selecting this model does not normalize data.

Approach 2 lists one row per model family with independent **Standard** and **Normalized** checkboxes. Select either variant or both for comparison. Results and reports identify each selected variant separately.

Response profile model notes

Finite-Time Power Decay uses the surface-erosion form based on the erodible slab, cylinder, and sphere model [4]. Logistic uses a sigmoidal transition to represent a delayed change [5]. Modified Gompertz adapts the form originally developed for microbial growth kinetics [6]; such functions have also described empirical polymer biodegradation weight-loss profiles [7]. The Linear + Power Decay form is an empirical option for remaining mass, strength, or another declining response. Exponential + Logistic combines an early exponential change with a delayed sigmoidal transition, allowing it to represent two-phase decay. Likewise, Logistic + Logistic combines two decreasing sigmoid components, where each transition has its own midpoint and width, and the response approaches zero at long times. These are empirical curve shapes rather than models of a device's chemistry or degradation mechanism. These forms can describe mass remaining, molecular weight, volume, or retained mechanical properties when the observed shape is appropriate. Burst and delayed/multiphase behavior in degradable polymer systems is discussed in the polymer erosion and PLGA release literature [8, 9, 10].

Each Approach 2 family has a less-flexible normalized variant. Exponential and Stretched Exponential fix amplitude at 1, while Double Exponential uses coefficients w and $1 - w$, fixing their sum at 1. Logistic and Modified Gompertz fix the baseline and total change at 1, giving asymptotes at 1 and 0. Finite-Time Power Decay, Linear + Power Decay, and Logistic + Logistic fix the response at time zero at 1. Exponential + Logistic fixes the baseline at 1 but retains separate fitted early and delayed magnitudes (B and D), so its terminal response is not fixed to zero. These constraints reduce the number of fitted parameters and assume that the analyzed response matches the fixed levels. Selecting a normalized model does not normalize the data.

4.4 Preprocessing options

Preprocessing prepares the uploaded data for comparison, fitting, and prediction. Normalization expresses response values relative to a reference value so datasets can be compared on a common relative scale. Scaling changes the numerical scale of time or response values without changing the underlying pattern. Interpolation is used when an approach requires paired in vitro and in vivo quantities that were not measured at exactly matching positions. The app constructs both response-at-time and time-at-response representations; these are used differently by Approaches 1 and 3 and are not used for Approach 2 fitting.

With supplied SD/SEM, leave **Log10 response scaling** unselected; the app rejects that combination even when responses are positive. The logarithmic fallback behavior described below applies only to otherwise supported settings.

4.4.1 Normalization

Option	Internal key	Description
Min-max normalization	min_max	Converts response values to a 0-1 scale using the observed minimum and maximum within each dataset. With supplied SD/SEM, the range must be finite and positive; errors are divided by that range, which is treated as fixed.
Initial-value normalization	initial_value	Divides each response by the first response value in that dataset when it is nonzero. Supplied SD/SEM is divided by its absolute value, which is treated as fixed. A zero initial response leaves values and errors unchanged.
Reverse response (1 - value)	one_minus	Applies 1 minus each response after any selected normalizations and before logarithmic scaling or interpolation. For data already expressed as fractions, this converts fraction lost to fraction remaining without requiring another normalization. The same selection applies to both calibration profiles and prediction inputs; values are not clipped to 0-1. Supplied SD/SEM is unchanged by this reversal.

4.4.2 Scaling

Option	Internal key	Description
Log10 time scaling	log_x	Uses a base-10 log scale for time values before fitting or interpolation. All analyzed times must be positive. The browser falls back to unscaled time when calibration times include zero or negative values and reports this in Data Processing. Prediction data must support the applied scaling; CLI inputs with invalid times are rejected.
Log10 response scaling	log_y	Uses a base-10 log scale for response values before fitting or interpolation. This option is unavailable with supplied SD/SEM. Without supplied uncertainty, all processed responses must be positive. The browser can fall back to unscaled response for invalid calibration values and reports this in Data Processing; prediction data must support the applied scaling, and CLI inputs with invalid responses are rejected.

4.4.3 Interpolation

Option	Internal key	Description
No interpolation	none	Uses only exact matches from the observed data. No response-at-time or time-at-response estimates are added, so unmatched positions remain unavailable for Approaches 1 and 3. Approach 2 uses observed points regardless of this setting.
Shared-grid interpolation	default	Uses linear interpolation in both directions: response values are estimated at the shared set of observed time points, and times are estimated at the shared set of observed response values. Estimates outside the measured overlap are unavailable. Approach 2 does not use these interpolated points for fitting.

Option	Internal key	Description
Densified interpolation	alternative	Adds intermediate grid points in both time and response-value directions before linear interpolation, producing denser paired mappings for Approaches 1 and 3. The added points are estimates rather than additional experimental observations. Approach 2 does not use them for fitting.

Reverse response $(1 - \text{value})$ can be used to transform fraction lost into fraction remaining. It is an optional checkbox applied after any selected normalizations and before scaling or interpolation. Use it for fraction-scale responses, applying normalization first if needed.

Supplied SD/SEM is rescaled automatically with the responses. When normalization divides responses by a constant, errors are divided by that constant's absolute value. The reference values used for initial-value and min-max normalization are treated as fixed; their own uncertainty is excluded. Response reversal leaves SD/SEM unchanged. Supply errors on the original response scale so the app applies each selected transformation once.

The app preserves observed time zero as $t = 0$ rather than replacing it with a small positive value. Because $\log_{10}(0)$ is undefined, Log10 time scaling requires positive times. If this scaling is selected and calibration times include zero or negative values, the browser analysis continues with unscaled time and displays a scaling-fallback warning in Data Processing. When Log10 time scaling is applied, prediction times must also be positive. Changing the time scale does not change response SD/SEM.

Without supplied uncertainty, Log10 response scaling requires positive processed responses. If this scaling is selected and calibration responses contain zero or negative values, the browser analysis continues with unscaled responses and reports the fallback. Prediction data must support the scaling applied to calibration. The preprocessing table reports the scaling actually applied. Model-specific numerical limitations at zero are handled during fitting rather than by changing the supplied times. For Approach 3, an in vivo/in vitro time ratio with an in vitro time of zero is undefined and is reported as unavailable, although a prediction observed at time zero can still map to time zero.

4.4.4 How interpolation is used by each approach

When interpolation is selected, the app constructs two complementary representations of the preprocessed fitting data:

- **Interpolation by time (response at shared times):** for a shared set of time points, the app estimates the corresponding in vitro and in vivo response values. This representation is used by Approach 3 for response-value mappings over time.
- **Interpolation by response value (time at shared values):** for a shared set of response values, the app estimates the corresponding in vitro and in vivo times. This representation is used by Approach 1 and by the time-mapping part of Approach 3.

The Data Processing results tab shows these two representations separately so that the points used for each mapping can be reviewed directly. With the shared-grid interpolation option, the shared grids are based on the union of observed time points and the union of observed response values, respectively. Densified interpolation adds intermediate grid points in both directions before interpolation. Estimates outside the overlapping data range are not used.

Approach 1: Direct Correlation. Approach 1 relates the in vitro and in vivo *times associated with the same response value*. If the two datasets contain exactly matching response values, those observed pairs can be used directly. Otherwise, interpolation by response value estimates the time at common response values so that the in vitro and in vivo times can be paired for fitting. Matching the original experimental time schedules is not required because the fitted relationship is in vivo time as a function of in vitro time at equivalent response states.

Approach 2: Independent Model Fitting. Approach 2 fits the selected model family independently to each preprocessed observed dataset. The in vitro and in vivo time points do not need to match, and the response values do not need to match, so no alignment by interpolation is required. The app therefore excludes interpolated points from Approach 2 fitting. Interpolated values are deterministic estimates from the measured observations rather than additional independent observations; treating them as fitting data could change the apparent sample size, fit metrics,

cross-validation inputs, residual degrees of freedom, and covariance-based parameter uncertainty without adding new experimental information. Normalization and scaling, when selected, are still applied before the independent fits.

Approach 3: Direct Mapping. Approach 3 does not fit a parametric model and instead uses both interpolation directions. Response values at shared times are used to calculate response-value ratios or mappings over time. Times at shared response values are used to calculate time ratios or mappings across the response range. Both representations are therefore relevant when Approach 3 is selected.

With **No interpolation**, Approaches 1 and 3 can use only the matching points that are actually present in both datasets. If the datasets have few or no matching times or response values, there may be too few points for the comparison. Interpolation can provide additional matched points.

Interpolation by response value assumes that each response value corresponds to a single time. If the response rises and falls over the analyzed range, the same response value may occur at more than one time, making this mapping unclear. Review the by-response-value interpolation plot to confirm the results appear consistent with the data before using this mapping.

4.5 Performance metrics

Performance metrics are summary measures that help describe how closely a fitted model matches the observed data (e.g., R^2), how large the prediction errors are (e.g., RMSE), and, for some metrics, how well a model balances fit quality with model complexity (e.g., information criteria). The following performance metrics are included in the app.

Metric	Internal key	Preferred direction	Description
R^2	r_squared	Higher is better	Coefficient of determination. Higher values indicate that the fitted curve follows more of the observed response pattern. This remains an unweighted descriptive score when SD/SEM is supplied.
Adjusted R^2	adjusted_r_squared	Higher is better	R^2 adjusted for the number of fitted parameters. It can decrease when extra parameters do not improve the fit enough to justify the added complexity.
MSE	mse	Lower is better	Average squared difference between observed and fitted values. Lower values indicate smaller model errors, but the squared units can be harder to interpret directly.
RMSE	rmse	Lower is better	Typical model error in the units of the fitted output on the analyzed scale. Lower values indicate predictions closer to the observations. This remains an unweighted descriptive score when SD/SEM is supplied.
MAE	mae	Lower is better	Average absolute model error in the units of the fitted output on the analyzed scale. Lower values indicate predictions closer to the observations.
NRMSE (mean)	mnrmse	Lower is better	RMSE divided by the mean of the observed analyzed outputs. Lower values indicate smaller relative error when the mean is positive and away from zero; near-zero or negative means make that interpretation unsuitable.

Metric	Internal key	Preferred direction	Description
NRMSE (std. dev.)	nrmse	Lower is better	RMSE divided by the standard deviation of the observed analyzed outputs. Lower values indicate smaller error relative to the trajectory's variation, not relative to supplied observation SD/SEM.
AIC	aic	Lower is better	Akaike Information Criterion. AIC compares fitted models by considering both fit quality and the number of fitted parameters. Among comparable fitted models, a lower AIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute AIC values, including negative values, are not standalone quality grades. Supplied-error fits use a normal likelihood for the supplied observation-error model on the analyzed scale.
AIC_c	aicc	Lower is better	Corrected Akaike Information Criterion. AIC_c is AIC with an added small-sample correction that becomes increasingly relevant when the number of fitted observations is limited relative to the number of fitted parameters. The app retains its conventional correction when SD/SEM is supplied; it is an approximation rather than an exact small-sample guarantee for every fitted model. Among comparable fitted models, a lower AIC_c value generally indicates better empirical support for a model that balances fit quality and complexity; AIC_c is undefined when there are too few observations relative to the number of parameters.
BIC	bic	Lower is better	Bayesian Information Criterion. BIC compares fitted models by considering both fit quality and the number of fitted parameters, usually applying a stronger penalty for extra parameters than AIC. Among comparable fitted models, a lower BIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute BIC values, including negative values, are not standalone quality grades. Supplied-error fits use a normal likelihood for the supplied observation-error model on the analyzed scale.

For additional background on common error metrics, scale-dependent accuracy measures, and adjusted R^2, see references [11, 12, 13].

AIC_c adds the app's conventional small-sample correction to AIC when the number of fitted observations permits it. Its justification depends on the model and error assumptions, including whether SD/SEM is supplied. It remains

available, but a small dataset alone does not establish that it is the preferred criterion. See Appendix A.5 for the convention and interpretation.

By default, the app selects R^2, RMSE, and AIC_c. This is a software starting configuration, not a universal or required metric set. R^2 describes agreement with the observed pattern, RMSE reports typical error in the units of the analyzed fitted output, and AIC_c summarizes relative support within the candidate set using a complexity penalty and the conventional correction. Additional metrics can provide context for the intended analysis.

4.6 Cross-validation options

Cross-validation provides an estimate of the model's ability to predict observations not used for fitting. The app does this by repeatedly setting aside part of the dataset, fitting the model to the remaining observations, and comparing model predictions with the observations that were set aside. When the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable.

In this app, CV is intended as a **prediction-error diagnostic** and sensitivity check, not as a confidence interval or parameter uncertainty estimate. If the error from the model fitted to the full dataset is similar to the CV error, that provides reassurance that the fitted relationship is not overly dependent on a particular subset of observations. If the CV error is much worse, the fitted relationship might be sensitive to the available data, the selected splitting scheme, or the number and placement of time points.

An optional **CV fit variability (mean ± SD)** plot provides a complementary view of that sensitivity. For every CV split that fits successfully, the app evaluates the model fitted to the training subset over the full displayed x-range. At each x-value, it calculates the mean fitted response and standard deviation across the CV fits. The plot shows the full-data model fitted to all observations, the mean CV-training fit, and a shaded mean ± one-standard-deviation band. A wider band indicates that the fitted curve changes more depending on which observations are included in the training data. This band summarizes variation among CV fits; it is not a confidence interval, prediction interval, or parameter-uncertainty interval. Its appearance depends on the selected CV splitting scheme and the number of successful CV fits.

For time-course data, the splitting scheme matters because nearby observations can contain related information, and early, middle, and late time points can represent different parts of the degradation process. Setting aside a random subset is a general check of whether the fitted curve captures the overall response pattern, but it is less time-aware. Setting aside one time point mainly asks whether the model can recover an individual omitted observation. Setting aside a contiguous block asks whether the model can bridge a missing region of the profile.

Shuffle split is the default CV scheme because it provides a general robustness check of whether the selected curve form captures the overall degradation or correlation pattern. It is not a time-aware forecasting test. Leave-contiguous-block-out may be informative when sensitivity to a missing time region is a primary concern; leave-one-timepoint-out may be informative when the dataset is too small to set aside larger groups of observations. When results change substantially across CV schemes, the fitted plots, residual diagnostics, parameter uncertainty, model complexity, and scientific plausibility provide additional context rather than treating CV as a pass/fail result.

The default CV settings reflect the tool developers' experience with representative IVIVC datasets and are not intended to be universally applicable. Alternative settings may be appropriate depending on the characteristics of the specific dataset and analysis goals.

For background on cross-validation procedures and their use in model assessment and selection, including issues that arise with time series, see references [14, 15, 16, 17].

4.6.1 Choosing a cross-validation scheme

The table below summarizes the schemes available in the app. The "CV error describes" column explains what kind of prediction sensitivity each scheme is most directly measuring.

Scheme	CV error describes	Useful when	Main caution
Shuffle split	Prediction error for repeated random subsets of observations set aside from fitting.	Default general robustness check for whether the fitted curve captures the overall response pattern.	Not time-aware; can appear overly reassuring when nearby time points carry similar information.
Leave-one-timepoint-out	Prediction error when one time point is omitted at a time; sensitivity to setting aside one observation.	Useful for very small datasets because each repeated fit still uses nearly all of the available data.	Can be a gentle test when neighboring time points have similar response values to the omitted point.
Leave-contiguous-block-out	Prediction error when a block of adjacent time points is omitted; sensitivity to a gap in the observed profile.	Primary time-aware sensitivity check, especially when a missing time region is a practical concern.	Larger blocks make the test harder but can leave too few points for the model to fit stably. Smaller blocks retain more observations for each repeated fit.
None	No cross-validation is performed; only the model fitted to the full dataset is summarized.	Useful when there are too few observations to split meaningfully, repeated fits are numerically unstable, or the goal is simply to examine the fitted model itself.	Removes one diagnostic tool. When CV is skipped, other outputs like fitted plots, residuals, full-fit metrics, parameter diagnostics, and scientific plausibility provide the remaining model-review context.

4.6.2 Cross-validation parameters

Parameter	Internal key	Description
Cross-validation scheme	cv_scheme	Selects the splitting method used to estimate cross-validation performance for fitted parametric models. Shuffle split is the default general robustness check.
CV splits/block size	cv_n_splits	For Shuffle split, the number of repeated random train/test splits. For Leave-contiguous-block-out, the number of adjacent time points set aside per split. The setup page does not enforce a maximum block size because the uploaded data are not read until Analyze is selected; larger blocks may leave too few observations for stable fitting. This setting is not used for Leave-one-timepoint-out or None.
Shuffle split test size	cv_test_size	The fraction of observations used as test data in each Shuffle split repeat. Smaller values leave more observations for fitting; larger values make the test more demanding but leave fewer observations for training.
CV random seed	cv_random_state	Seed used to make randomized cross-validation splits reproducible. This applies to Shuffle split.

4.7 Grid-search initialization

When the app fits a mathematical curve, it needs starting values for the model parameters. Poor starting values can lead to an unstable or unreasonable fit, especially for nonlinear models. Grid-search initialization reduces this risk by trying many candidate starting-value sets before nonlinear least-squares fitting.

The app evaluates those candidate starts, chooses the most promising one, and then runs the final curve-fitting step from that starting point. This improves robustness, but it does not establish that the fit is the global optimum.

The app provides default grid-search settings as its initial configuration. Options for addressing unstable fitting include increasing the number of random starts, adjusting the candidate starting-value range, changing the random seed, or evaluating whether the selected model form describes the data adequately.

Parameter	Internal key	Description
Grid-search random starts	grid_search_num_points	Number of candidate starting parameter sets evaluated before nonlinear least-squares fitting. Higher values may improve robustness but increase runtime.
Grid-search CPU cores	grid_search_num_cores	Number of CPU cores used for the starting-point search. Only available via command-line interface.
Grid-search parameter minimum	grid_search_param_min	Lower bound for randomly sampled candidate starting parameter values.
Grid-search parameter maximum	grid_search_param_max	Upper bound for randomly sampled candidate starting parameter values.
Grid-search random seed	grid_search_random_state	Seed used to make randomized starting-point sampling reproducible. Changing the seed changes the sampled candidates while keeping the same bounds and number of starts.

4.8 Diagnostic plot display

The CV fit variability plot, raw residual plots, and normal quantile-quantile (Q-Q) plots are not selected by default in the Advanced Analysis section to produce a more compact results view. These plots can be selected to analyze the stability of CV and the residual behavior. The CV fit variability plot is available only when cross-validation is performed. It is placed below the goodness-of-fit table for the corresponding fitted dataset.

5 Outputs

The IVIVC app generates interactive results and downloadable reports that summarize fitted model behavior, model-comparison metrics, diagnostic information, and user-selected analysis settings. Output availability depends on the selected approaches, whether fitting succeeds, and whether a prediction dataset was supplied.

Goodness of fit describes agreement with the data used for fitting. Cross-validation summarizes prediction error and sensitivity to observations set aside under the selected splitting scheme. Independent (external) validation uses additional paired in vitro and in vivo experimental data that were not used to develop or select the relationship; an in vitro prediction dataset alone does not provide this independent comparison. Uncertainty displays have different meanings, summarized in Section 5.3.

Output	Description
Preprocessing outputs	Processed data tables and plots showing the data used for fitting and prediction. When uncertainty is supplied, the record includes entered and analyzed SD/SEM aligned by source row.
Model fitting outputs	Parameter estimates and approximate confidence intervals, fitted curves with parameter-uncertainty bands, supplied SD/SEM bars for Approach 2 observations, selected metrics, and optional CV fit-variability and residual diagnostic plots.

Output	Description
Model comparison outputs	Tables comparing selected approaches, models, metrics, relative model evidence, and cross-validation summaries.
Prediction outputs	Predicted in vivo time or response, depending on the method, with available interval bounds, calibration-range cautions, and calculation status in plots and downloadable tables.
Selected model & summary of outputs	User-selected report summaries and optional report export for incorporation into external documentation.

Review Data Processing to confirm which observations and errors were used, especially after normalization or removal of missing data. Entered and analyzed errors are reported separately so their rescaling can be checked. Plot-data downloads and reports retain the calculated interval endpoints and reasons for unavailable or extrapolated results.

5.1 Relative model evidence

Relative model evidence tables compare candidate parametric models using information criteria (i.e., AIC, AIC_c, and/or BIC), showing which models have more support from the observed data after balancing fit quality against model complexity. These comparisons are only meaningful among models fitted to the same data, with the same preprocessing and uncertainty assumptions, and within the same IVIVC approach.

These tables are shown only when one or more information criteria are selected as metrics. Each selected criterion is interpreted separately because AIC, AIC_c, and BIC can rank the same models differently. Approach 2 provides separate in vitro and in vivo summaries and combined evidence where the required values are available. With supplied errors, the criteria reflect that observation-error model; AIC_c retains the app's conventional correction. See Appendix A for interpretation guidance and limitations.

The relative model evidence tables contain the following quantities.

Table output	What it means
n	Number of observations used to fit the model.
k	Number of fitted model parameters. Models with more parameters can fit more flexible shapes, so information criteria include a complexity penalty.
AIC, AIC_c, or BIC	Information-criterion value for the fitted model. Lower values indicate stronger relative support within the candidate set.
Delta	Difference between a model's criterion value and the lowest value in the candidate set. Values near 0 indicate similar support to the leading model; larger values indicate less support.
Weight	Normalized relative-support score across the candidate set. These comparative summaries are not absolute probabilities that a model is correct and differ from the SD/SEM-based observation weights used in fitting.
Evidence ratio	Ratio of the highest weight in the candidate set to the model's weight. Larger values indicate less relative support compared with the leading model.
Support badge	Rule-of-thumb label based on the delta value. Intended to aid user review, not to recommend a model.

5.2 Prediction outputs

The optional prediction dataset contains new in vitro time and response values. After preprocessing, the app applies the fitted relationship using the prediction time, response, or both. Some methods predict the corresponding **in vivo time**, while others predict the corresponding **in vivo response value**.

The file must contain both columns. The table below identifies which columns enter each calculation; the other coordinate may still be carried into the plot. These methods answer different prediction questions and should not be treated as interchangeable outputs.

Approach / method	Prediction columns used	How the prediction is made	Output
Approach 1: Direct correlation	In vitro time	Applies the fitted relationship between in vitro time and in vivo time: $t_{\text{vivo,pred}} = f(t_{\text{vitro,pred}})$. The prediction response value is not used in this calculation.	In vivo time
Approach 2: Response-value ratio	In vitro time and response	Evaluates the fitted in vitro and in vivo curves at the prediction time, calculates their response ratio, and uses that ratio to scale the observed prediction response: $y_{\text{vivo,pred}} = y_{\text{vitro,pred}}\, f_{\text{vivo}}(t)/f_{\text{vitro}}(t)$.	In vivo response
Approach 2: Time-constant ratio	In vitro time	Scales the prediction time by the ratio of the fitted in vivo and in vitro time constants: $t_{\text{vivo,pred}} = t_{\text{vitro,pred}}(\tau_{\text{vivo}}/\tau_{\text{vitro}})$. The prediction response value is not used in this calculation.	In vivo time
Approach 2: Observed-response inverse mapping	In vitro response	Finds the time at which the fitted in vivo curve reaches the observed prediction response: $t_{\text{vivo,pred}} = f_{\text{vivo}}^{-1}(y_{\text{vitro,pred}})$. The prediction time is not used to calculate the mapped time.	In vivo time
Approach 3: Response-value scaling	In vitro time and response	Uses the fitting datasets to estimate the in vivo/in vitro response ratio at the prediction time, then multiplies the observed prediction response by that ratio.	In vivo response
Approach 3: Time scaling	In vitro time and response	Uses the fitting datasets to estimate the in vivo/in vitro time ratio associated with the prediction response, then multiplies the prediction time by that ratio.	In vivo time

The three Approach 2 methods emphasize different features of the profiles. Response-value-ratio rescaling preserves time-dependent differences between the fitted curves but can become unstable near a zero fitted in vitro response. A zero or nonfinite fitted denominator makes the point unavailable; if the denominator's uncertainty interval includes zero, the prediction interval is unavailable. Time-constant-ratio rescaling provides one overall time scale and can miss differences in curve shape; its time constants also depend on fitted behavior beyond the observed times. Observed-response inverse mapping treats the supplied response as a degradation state.

Uncertainty in Approach 2 predictions. Calibration uncertainty comes from the relevant fitted curves. A source with supplied SD/SEM uses that error scale; a source without supplied errors uses scatter around its fitted curve. Without prediction-input SD/SEM, the prediction responses are treated as fixed. The table below shows how supplying prediction-response uncertainty affects the approximate 95% intervals before adding optional future measurement variability.

Approach 2 method	Calibration contribution	Effect of prediction-response SD/SEM
Response-value ratio	Both fitted curves contribute to the predicted response's vertical bounds.	Combines with calibration uncertainty in those vertical bounds. Prediction time is fixed.
Time-constant ratio	Both fitted curves contribute to the predicted time's horizontal bounds.	Adds vertical input-response bounds; it does not change the mapped time or its horizontal bounds.

Approach 2 method	Calibration contribution	Effect of prediction-response SD/SEM
Observed-response inverse mapping	The fitted in vivo curve contributes to the predicted time's horizontal bounds.	Propagates together with in vivo parameter uncertainty into the time bounds; vertical bounds show input-response uncertainty alone.

For time-constant and inverse predictions, the vertical center is the supplied in vitro response retained by the mapping. Without future measurement variability, its bounds describe that input response only.

Point estimates use the analyzed inputs and fitted parameters. Prediction-input SD/SEM enters the interval calculation without changing the point estimate when the calibration fits are held fixed. Supplying calibration errors can change those fits and therefore the point estimates.

When future variability is selected, vertical bounds combine propagated response uncertainty with independent normal future measurement scatter. They are approximate 95% prediction intervals for one future in vivo measurement, or for a future mean when SEM is selected. For time-constant and inverse mapping, these future responses accompany the mapped prediction; horizontal mapped-time intervals keep their existing meaning. Reusing source uncertainty explicitly assumes comparable future precision; it does not remove the source's contribution to propagation. Calibration reuse selects the nearest time on the analyzed scale (the earlier time for a tie), holds that SD/SEM fixed during propagation, and uses the endpoint error outside calibration. Downloads and reports record the source, applied errors, and interval status.

Prediction ranges and unavailable results. Interpolation estimates between calibration observations; extrapolation extends a relationship beyond its calibration range. Approach 1 checks the paired-time ranges used for fitting. Approach 2 response-value ratios check the times at which both fitted curves are evaluated, while time-constant scaling checks the input and mapped times. These parametric methods can produce predictions beyond the observed ranges; narrow intervals do not establish that those predictions are reliable.

Observed-response inverse mapping checks the in vivo calibration response and time ranges. Its numerical search can extend beyond calibration without clipping responses to the calibration endpoints. Flat or nonunique responses, nonmonotonic continuations, invalid model domains, or unresolved roots can make a prediction unavailable. An unavailable result means a reliable unique time was not calculated; it does not establish that no mathematical solution exists.

Approach 3 obtains scaling factors only within the supported calibration time or response range. The resulting scaled in vivo response or time can still lie outside the observed in vivo range.

An **Extrapolation** caution identifies results outside the relevant calibration ranges. An interval can carry a separate caution even when its point estimate is inside calibration. Review prediction status and interval status separately: a point may be available without its interval, and one prediction method may be unavailable while another produces a result. Downloads and reports retain the reasons and calibration ranges, including failed-draw information for inverse intervals.

All predictions, interval bounds, and calibration ranges use the analyzed, preprocessed coordinates. The app does not automatically undo the selected normalization, response reversal, or logarithmic scaling. See Section 5.3 for the interpretation and limitations of the displayed intervals.

5.3 Uncertainty and error bars

The app displays several uncertainty summaries. Check which quantity a bar, band, or plus-minus value describes before comparing their widths.

Display	What it means
Observed-data error bars	Approach 2 calibration observations with supplied uncertainty show ± one SD or SEM. These are the supplied errors, not 95% intervals or uncertainty bands for the fitted curve.

Display	What it means
Fitted-curve shading	Approximate pointwise 95% uncertainty in the fitted curve because its parameters were estimated from data.
Prediction error bars or interval segments	Approximate pointwise 95% bounds for the predicted quantity, propagating calibration uncertainty and, for Approach 2 when supplied, prediction-response uncertainty as described above. Time-prediction plots may also show separate input-response bounds. Selecting future variability adds future measurement scatter to the vertical response bounds.
Parameter and time-constant tables	Estimate ± the half-width of an approximate 95% interval for an individual coefficient or time constant, where available. The plus-minus value is not the supplied observation SD/SEM.
CV fit variability shading	Mean ± one SD across successful CV-training fits. This describes sensitivity to omitted observations; it is not a confidence or prediction interval.

"Pointwise" means that a curve or prediction interval applies at each displayed position, not to the whole curve simultaneously. When both axes have bounds, each describes its coordinate separately; the resulting rectangle is not a joint 95% region.

Fitted-curve and prediction intervals propagate parameter uncertainty while retaining estimated relationships among parameters within each fit. Separate calibration fits are sampled independently. Supplied prediction-response errors are modeled as independent normal errors across entered rows. Time coordinates, normalization constants, the selected model, and preprocessing choices are held fixed. Supplied SD/SEM values are treated as given. Future in vivo measurement variability is included only when explicitly selected for Approach 2 prediction responses. Otherwise, these are propagated-uncertainty intervals rather than intervals for future observations. Extra residual scatter, uncertainty about model choice, and model inadequacy are excluded in either case.

Approach 1 displays uncertainty in its fitted time relationship conditional on the paired times used for fitting; supplied measurement-error propagation is available only for Approach 2. Approach 3 calculates deterministic mappings without formal uncertainty intervals. Missing error bars do not mean a mapping is exact.

A valid central estimate can remain available when its interval cannot be calculated. An unavailable interval does not mean zero uncertainty. Inverse time intervals and their optional future-response intervals require the configured minimum fraction of requested draws to map successfully at that point. If some draws fail, the displayed interval is labeled conditional and encloses the middle 95% of successful mappings; it does not describe all requested outcomes. Successful/requested counts and failure reasons are retained. Other sampled intervals require every requested calculation to succeed. Some intervals from sampled predictions do not contain the point estimate; the app draws them as separate segments at their calculated endpoints. Sampling settings and calculation-status details are recorded with the prediction results. See Appendix A for further interpretation of fitted uncertainty and diagnostics.

These intervals are not intended to define acceptability thresholds for regulatory submissions. Applicable credibility and evidence considerations depend on the intended use and applicable FDA guidance [1, 2].

Very large uncertainty bands can extend beyond the initial plot range and remain out of view until the plot is rescaled. When this occurs, the app displays an informational badge near the affected fitting plot. A separate warning is shown if one or more uncertainty bounds are nonfinite or the uncertainty band cannot be calculated. The badge describes the display condition but does not identify its cause; the fitted parameter diagnostics, model form, and available data provide context for interpreting it.

6 Miscellaneous

6.1 Software verification and validation

Software verification concerns whether the app works as described and performs its calculations correctly. Software validation concerns whether it supports its intended empirical correlation tasks. Sponsors using the app as provided do not need to verify or validate its software functionality. These questions are separate from assessing whether an individual IVIVC is suitable for its intended use.

6.2 Command-line interface

Run `cli.py` from the repository root with the app's Python environment activated. The CLI supports Approach 1 or 2, with one approach and one model per invocation. Use internal model and metric keys; repeat `--metric` for additional metrics. Run `python cli.py --help` to see the available choices.

CSV and Excel files use the same first-four/first-two column conventions as the browser. Files or quoted glob patterns, such as `data/*.csv`, can be supplied. Excel fitting files require `--sheet`; selectors use zero-based indices and support negative indices, lists, and Python-style slices. For example, `--sheet 0` selects the first sheet and `--sheet 0,2` selects the first and third. Excel prediction files require `--prediction-sheet` selecting exactly one sheet. The same settings and prediction dataset apply to all selected fitting files and sheets.

Basic fitting example. Assume `calibration.csv` contains fraction-remaining responses compatible with the normalized exponential model. The examples use one CPU core and disable CV to keep the illustration simple; choose settings appropriate to the analysis. Backslashes continue commands across lines in a shell such as Bash; enter the command on one line if the shell uses different continuation syntax.

```
python cli.py calibration.csv --approach approach2 \
  --model exponential_normalized --metric rmse --metric aicc \
  --cv-scheme none --grid-search-num-cores 1 --out fits.csv
```

Uncertainty options. Use the prefixes `in-vitro`, `in-vivo`, and `prediction` for the three datasets. Replace `PREFIX` in the flags below with the relevant prefix. The SD/SEM interpretation, original response units, and count rules in Section 4.2 apply to the CLI as well.

Flag	Meaning
`--PREFIX-uncertainty-mode`	`none` (default), `constant`, or `column`.
`--PREFIX-uncertainty-value`	Positive absolute SD/SEM for `constant` mode.
`--PREFIX-uncertainty-column`	Error-column header for `column` mode.
`--PREFIX-response-kind`	`individual` (default) uses SD; `mean` uses SEM directly.
`--PREFIX-replicate-column`	Optional positive-integer count column for means; metadata only.

Any supplied uncertainty requires `--approach approach2` and `--independent-errors` to confirm the assumptions. Leave `--scaling log_y` unselected with supplied errors. CLI preprocessing rejects invalid logarithmic inputs rather than using the browser's calibration fallback.

For future-response prediction intervals, add `--future-uncertainty-mode` with `constant`, `prediction`, or `calibration`; `none` is the default. Constant mode also takes `--future-uncertainty-value` and `--future-response-kind individual` (SD, the default) or `mean` (SEM). The value uses the original in vivo calibration response units. Reuse modes require the corresponding supplied source errors. Future variability requires `--prediction`, Approach 2 alone, and `--independent-errors`.

Fitting and prediction with SEM. For this example, both files contain reported means on the same fraction-remaining scale. After its four required columns, `calibration.csv` has `vitro_sem` and count column `vitro_n`; `prediction.csv` has its two required columns plus `prediction_sem`. The constant in vivo SEM of 0.03 is illustrative; use errors appropriate to the data.

```
python cli.py calibration.csv --approach approach2 \
  --model exponential_normalized --metric rmse --metric aicc \
  --in-vitro-response-kind mean \
  --in-vitro-uncertainty-mode column \
  --in-vitro-uncertainty-column vitro_sem \
  --in-vitro-replicate-column vitro_n \
  --in-vivo-response-kind mean \
  --in-vivo-uncertainty-mode constant \
  --in-vivo-uncertainty-value 0.03 \
  --prediction prediction.csv --prediction-response-kind mean \
  --prediction-uncertainty-mode column \
  --prediction-uncertainty-column prediction_sem \
```

```
    --independent-errors --cv-scheme none \
    --grid-search-num-cores 1 \
    --out fits.csv --prediction-out predictions.csv
```

`--out` writes fit-summary rows; `--prediction-out` writes separate prediction rows identified by method. If an output path is omitted, that result is printed to the terminal. Prediction outputs require `--prediction`; valid fitting summaries can remain available if prediction fails, with the failure recorded in the prediction output. With supplied calibration errors, the fit CSV retains entered and analyzed error values, covariance conventions, and normalization records; arrays and records use JSON in individual cells. Prediction rows retain supplied input errors, available bounds, statuses, and the relevant uncertainty assumptions, including future variability and propagated-only response bounds when selected.

6.3 Extending the app

The IVIVC app is designed so that common extensions can be added with minimal changes to the main application code. In many cases, new models, performance metrics, and preprocessing methods can be added by following the structure of the existing files in the relevant subdirectories.

6.4 Data handling

The app does not save uploaded datasets or fitting results to persistent storage. Uploaded data are processed for the current analysis, and generated results and downloads are returned directly to the user. When running the app locally, no data leaves the user's computer.

Appendices

A How to Interpret Model Performance

No single metric determines whether an IVIVC model is meaningful. An evaluation of model performance can include goodness of fit, prediction error, model complexity, cross-validation behavior, residual patterns, parameter uncertainty, and whether the fitted relationship is scientifically plausible for the material, degradation mechanism, and intended use. Performance metrics are descriptive and comparative tools, not acceptance criteria. Small datasets can produce unstable estimates, and similar metric values do not necessarily indicate equivalent scientific validity. For analyses intended to support regulatory submissions, the depth of supporting evidence depends on the model's intended use and applicable credibility considerations [1, 2].

The app reports metrics and comparison tables as descriptive aids for user review; the scope and advisory status described in the Introduction apply throughout this appendix. Users are responsible for determining how to document preprocessing assumptions, fitting settings, model-selection rationale, limitations, and any planned use of the model outside the fitted data range, consistent with the intended use and applicable credibility considerations.

A.1 Measurement uncertainty and weighted fitting

For Approach 2, supplied SD or SEM represents the standard uncertainty σ_i of the response entered at row i. SD describes variation among individual measurements; SEM describes uncertainty in a reported mean. These quantities differ from a fitted parameter's standard error, which is calculated from the parameter covariance matrix. A mean is one fitted observation, even if it summarizes several replicates. Optional replicate counts are metadata: they do not increase the fitted sample size or divide an entered SEM again. Input choices are described in Section 4.2.

With supplied errors, the app minimizes squared residuals relative to their SD/SEM:

$$Q = \sum_{i=1}^{n} \left(\frac{r_i}{\sigma_i} \right)^2, \qquad r_i = y_i - f(t_i; \hat{\theta}), \qquad w_i = \frac{1}{\sigma_i^2}.$$

Here r_i is observed minus fitted response and w_i is an observation weight. The responses and errors use the analyzed scale after preprocessing. Smaller errors give an observation more influence. The same errors are used in the starting-value search, the full fit, and the corresponding CV training fits. This is weighted least squares [18].

The app treats the supplied error scale as absolute: the fitted covariance is not rescaled to match the residual scatter. Without supplied errors, each calibration source retains ordinary least squares and a covariance scale estimated from its residuals [19]. Equal constant errors give the ordinary least-squares objective up to a constant factor, but the covariance conventions can still differ. A perfect fit need not have zero parameter uncertainty when positive SD/SEM is supplied.

Supplied errors are treated as given, even when they were estimated from few replicates. The app does not propagate uncertainty in those error estimates. It assumes independent measurement errors across entered rows and times, with prediction measurements independent of calibration. Selecting a different CV scheme does not introduce a correlated-error model.

Normalization rescales the response errors using the selected transformation, with reference values treated as fixed. It does not propagate the joint uncertainty involved in estimating a normalization reference from the same observations. Normalizing a response to 1 therefore does not make its supplied SD/SEM zero. These are conditional assumptions, not evidence that normalization references or measurements are exact.

A.2 Model comparability before interpreting metrics

Performance metrics are most meaningful when the models being compared were evaluated under identical conditions. Comparisons should use the same observations, preprocessing, analyzed output scale, selected IVIVC approach, and source-specific uncertainty assumptions.

In particular, do not compare AIC, AIC_c, or BIC across runs with different data, response transformations, or entered SD/SEM. These criteria describe relative support within a comparable candidate set, not absolute model quality. Approach 3 uses direct mappings rather than a parametric likelihood fit and does not have the same information-criterion comparison.

A.3 Metric ranges, direction, and interpretation

Metric	Range / direction	Interpretation guidance
R^2	Best possible value is 1. Values near 0 indicate little improvement over predicting the mean response. Can be negative if the model performs worse than the mean-response baseline. Higher is better.	Proportion of response variance explained by the model [13]. High R^2 does not establish that the model is mechanistic, causal, predictive outside the fitted data range, or adequately supported for the intended use.
Adjusted R^2	Higher is better. Can be lower than R^2.	Penalizes additional fitted parameters, making it more informative than R^2 when comparing models of different complexity [13].
MAE	Lower is better. Units match the analyzed fitted output.	Mean absolute error [11]. Interpret relative to assay variability, measurement uncertainty, and the intended use.
MSE	Lower is better. Units are squared units of the analyzed fitted output.	Mean squared prediction error [11]. Useful computationally, but less intuitive than RMSE because the units are squared.
RMSE	Lower is better. Units match the analyzed fitted output.	Root mean squared error, a summary of typical prediction error [11]. Penalizes larger errors more heavily than MAE.
NRMSE	Lower is better when the denominator is positive.	RMSE divided by the observed output SD (*std. dev.*) or mean (*mean*), as identified in the app. Near-zero denominators make the result unstable; a zero denominator makes it undefined.
AIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity [20, 21, 22]. A lower value indicates a model that explains the data well without unnecessary parameters. Absolute values are not standalone quality grades.
AIC_c	Lower indicates more support within a comparable set. The app requires $n > k + 1$.	AIC plus the app's conventional small-sample correction. Its justification depends on the model and error assumptions; see the qualification below.
BIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity, typically applying a stronger penalty for additional parameters than AIC [23, 21, 22].
Cross-validation metrics	Direction follows the underlying metric: higher is better for R^2-type metrics, lower is better for error metrics.	Estimate prediction performance under the selected splitting scheme [14, 15, 16, 17]. Can help identify overfitting, instability, or sensitivity to individual observations or data splits.
Fit/CV ratio	Values near 1 indicate similar fitted and cross-validation performance.	A ratio near 1 suggests similar full-fit and cross-validation behavior. Substantial deviations can indicate overfitting, instability, or sensitivity to the splitting scheme.

A.4 Interpreting goodness of fit and prediction error

Goodness-of-fit metrics—R^2, adjusted R^2, MSE, MAE, RMSE, and NRMSE—describe how well the model fits the data used to estimate its parameters. Their interpretation depends on practical context, not just numerical value. The app does not apply a universal IVIVC performance cutoff; interpretation depends on the intended use and, for regulatory contexts, applicable FDA guidance [1, 2].

Strong goodness-of-fit can still be misleading, particularly for small datasets or models with many parameters.

Favorable fit metrics do not rule out unstable parameters, poor cross-validation performance, systematic residual patterns, or a fitted curve shape that is not scientifically plausible for the degradation process being modeled.

The ordinary descriptive metrics remain unweighted when SD/SEM is supplied. A weighted fit can therefore have a larger ordinary RMSE than an unweighted fit while better matching observations with smaller supplied errors. Review these scores alongside the stated fitting objective rather than expecting every metric to improve.

RMSE uses the units of the analyzed fitted output: in vivo time for Approach 1, or response for Approach 2. The two NRMSE options change the denominator rather than incorporating experimental error bars [12]:

- **NRMSE (mean)** asks how large the typical error is relative to the average analyzed output. It is most interpretable for a positive mean away from zero; a negative mean does not give the usual positive relative-error interpretation.
- **NRMSE (std. dev.)** asks how large the typical error is relative to the spread of the observed outputs. That spread includes the changing trajectory across time; it is not the SD/SEM supplied for an individual observation or mean.

NRMSE values are not directly comparable across datasets or response scales when the normalization method differs. More broadly, the practical importance of a given error level depends on context: a numerically small error may still matter when the relevant degradation endpoint has narrow variation, while a larger error may be less consequential during exploratory model development when high predictive precision is not central to the analysis.

A.5 Interpreting relative model evidence

Relative model evidence tables compare candidate models using information criteria (i.e., AIC, AIC_c, and BIC), which balance goodness of fit against model complexity. In this context, *empirical support* means that the observed data provide more support for one model than another within the displayed candidate set, based on the selected information criterion. It does not establish that a model is correct, mechanistic, or predictive outside the observed range. Whether the evidence base is sufficient to support a regulatory submission depends on the intended use and applicable FDA guidance [1, 2].

How supplied errors enter the criteria

For a calibration source with supplied errors, the app uses an independent normal observation-error model. At the fitted parameters, its negative twice log likelihood is

$$D = -2\log L = \sum_{i=1}^{n}\left[\left(\frac{r_i}{\sigma_i}\right)^2 + \log(2\pi\sigma_i^2)\right], \qquad \text{AIC} = D + 2k, \quad \text{BIC} = D + k\log n.$$

Here n counts the response rows used in that fit and k counts fitted curve parameters. Replicate counts do not increase n, and fixed model coefficients are not included in k.

Without supplied errors, the app retains its RSS-based criterion convention, using $n\log(\text{RSS}/n)$ in place of D, with parameter-independent constants omitted and a numerical floor near zero RSS. This estimates a common error scale from the residuals. It answers a different error-model question from treating entered SD/SEM as an absolute scale. Raw RSS alone therefore does not describe the supplied-error likelihood. In both paths the app counts curve parameters only; it does not add an estimated error-scale parameter to k. These statements document the software's convention.

AIC_c: retained correction and its interpretation

For either error convention, the app retains

$$\text{AIC}_\text{c} = \text{AIC} + \frac{2k(k+1)}{n-k-1}, \qquad n > k+1.$$

When that sample-size condition fails, AIC_c is unavailable even if AIC or BIC can be calculated. A low observation-to-parameter ratio makes small-sample assumptions important; the often-cited $n/k < 40$ rule is a regression rule of thumb [21], not an automatic reason to prefer this correction for every fit.

Small-sample corrections depend on the statistical setting [24]. Known and estimated error variances are different settings, as illustrated for normal linear regression by Maier's analysis (a preprint) [25]. The app keeps conventional AIC_c available with supplied SD/SEM, but does not claim that it is an exact small-sample correction for every nonlinear model and supplied-error setting. Interpret it alongside the other selected criteria and model diagnostics, using the same uncertainty assumptions across candidates.

Separate and combined Approach 2 support

Approach 2 reports in vitro support, in vivo support, and a combined summary. The combined value for each criterion is the sum of its two source-specific values, under the independent-source fitting assumptions. In particular, combined AIC_c adds the two separate corrections and requires both to be defined. It is not a new correction calculated from pooled n and k; combined BIC likewise adds the separate BIC values.

One source may use supplied errors while the other estimates its scale from residuals. That mixed configuration remains available, provided each source's observations and error assumptions are kept the same across the compared models. Separate and combined support can differ because a model may describe one source more strongly than the other.

Rules of thumb for delta values

Delta is the difference between a model's criterion value and the lowest value in the candidate set. Common interpretive conventions are:

- **AIC / AIC_c**: delta 0–2 suggests similar support; 4–7 suggests less support; greater than 10 suggests essentially no support relative to the leading model [21].
- **BIC**: delta 0–2 suggests similar support; 2–6 suggests less support; 6–10 suggests substantially less support; greater than 10 suggests essentially no support relative to the leading model [26].

These are interpretive conventions, not regulatory acceptance criteria.

Evidence weights

For a given criterion, the evidence weight for model m is:

$$W_m = \frac{\exp\left(-\frac{1}{2}\Delta_m\right)}{\sum_{j=1}^{J}\exp\left(-\frac{1}{2}\Delta_j\right)} \tag{1}$$

where Δ_m is the delta value for model m and J is the number of candidate models. Larger weights indicate stronger relative support within the displayed set [21, 22]. These model-support weights differ from the observation weights $1/\sigma_i^2$ used during fitting and are not absolute probabilities that a model is correct.

When multiple models have similar support

If several models have similar delta values or weights, additional considerations can include model simplicity, parameter plausibility, residual behavior, cross-validation performance, and scientific interpretability.

A.6 Interpreting cross-validation performance

Cross-validation evaluates prediction performance using observations left out when the model was fit. In this app, it is most useful as a **sensitivity diagnostic**: when the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable regardless of which observations are set aside. When the two differ substantially, the model may be more sensitive to which specific data points were included in fitting, or to the particular way the data were split. Cross-validation does not measure uncertainty in the model parameters. The practical question to keep in mind is not just "how good is the score?" but also "what kind of prediction task did the cross-validation scheme actually create?"

What each scheme is testing

Different schemes create different prediction tasks. Shuffle split, the default, sets aside randomly selected observation rows and checks overall curve-shape robustness. Leave-one-timepoint-out omits one row at a time; observations sharing a time are not grouped together. Leave-contiguous-block-out omits adjacent rows in fitting-x order and checks sensitivity to a missing region. These checks can be gentle when nearby observations carry similar information.

Preprocessing and alignment are established before the splits, rather than rebuilt using each training subset alone. Starting values can also come from the full-data initialization search. The results therefore describe sensitivity within the chosen analysis setup; they are not independent validation of the complete analysis procedure.

These distinctions matter because adjacent time points in a degradation profile are often similar to one another, measurement errors may be related across time, and early, middle, and late portions of a degradation profile may reflect qualitatively different behavior. Standard cross-validation procedures are reviewed in [14]; additional considerations for temporally ordered or dependent data are discussed in [15, 16, 17].

Interpreting results for small IVIVC datasets

For the small datasets typical in IVIVC development, cross-validation results are intended as an exploratory diagnostic or sensitivity analysis rather than a definitive performance benchmark. Interpretation can include comparison of the CV metric with the corresponding metric for the model fitted to the full dataset together with whether the chosen splitting pattern reflects the practical prediction question of interest.

For example, strong leave-one-timepoint-out performance alongside weaker leave-contiguous-block-out performance may indicate that the model can interpolate individual missing points well, but becomes less reliable when an entire local region of the profile is absent from fitting. The importance of that distinction depends on the intended use of the model. For credibility assessment in the context of regulatory submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

CV fit variability plots

Average CV metrics summarize prediction error across the observations set aside and the CV splits, but do not show how much the fitted curve itself changes between training subsets. Two models can have similar average CV errors while differing in sensitivity to omitted observations.

The optional CV fit variability plot provides this complementary information by showing the full-data fit together with the mean fitted curve and mean $\pm$ one standard deviation across the successful CV training-set fits. A narrow band indicates that the fitted relationship changes relatively little across the CV splits. A wider band, or widening confined to a particular part of the curve, identifies regions where the fitted relationship is more sensitive to the training data, suggesting that additional data in that region may be particularly informative. A noticeable difference between the mean CV fit and the full-data fit can likewise indicate that repeatedly leaving out observations changes the fitted relationship in a systematic way.

The band is a descriptive sensitivity diagnostic, not a confidence interval, prediction interval, or parameter-uncertainty interval. Its appearance depends on the selected CV scheme and the number of successful CV fits; interpretation therefore also depends on the average CV metrics and the prediction task represented by the chosen splitting scheme.

Fit/CV ratios

The displayed ratio is the full-fit value divided by the mean CV value. For a positive error metric such as RMSE, worse CV error gives a ratio below 1. For positive R^2 values, a smaller CV score gives a ratio above 1. Ratios can be undefined or difficult to interpret near zero or with negative scores. Review the raw values, splitting scheme, and fitted plots; a ratio alone does not establish generalizability or overfitting.

Cross-validation comparison badges

The app displays cross-validation comparison badges as interpretive aids. Badges are assigned based on the preferred direction of the metric: for lower-is-better metrics such as RMSE, a worse CV result means the CV value is larger

than the full-fit value; for higher-is-better metrics such as R^2, a worse CV result means the CV value is smaller. The thresholds used are:

- **Similar**: CV performance is no more than 1.25-fold worse than the full-fit value, or is better under that metric.
- **Moderately worse**: CV performance is 1.25- to 2-fold worse than the full-fit value.
- **Substantially worse**: CV performance is more than 2-fold worse than the full-fit value.

The badges use a direction-aware worsening factor, rather than applying these thresholds directly to the displayed fit/CV ratio. Undefined factors produce an unavailable badge. These thresholds are heuristic review aids, not statistical acceptance criteria or regulatory standards.

Additional CV summaries with supplied SD/SEM

Each Approach 2 CV training fit uses the supplied errors for its training rows. The additional supplied-error summary compares predictions with the errors assigned to the omitted observations. If z_a is the omitted prediction error divided by that observation's SD/SEM, the pooled score is

$$\mathrm{RMSE}_{\mathrm{standardized}} = \sqrt{\frac{1}{M}\sum_{a=1}^{M} z_a^2}.$$

Here M counts test-observation occurrences across splits: a row used as test data repeatedly is counted repeatedly. It is not a count of additional independent measurements. A score near 1 means the omitted errors are comparable to the supplied SD/SEM scale on average, not that the model passes a calibrated test.

The observation negative log likelihood uses the supplied normal error model evaluated at each CV fitted prediction; the report also gives its mean per test occurrence. It assesses the fitted prediction against that measurement-error model without adding uncertainty from estimating the training-fit parameters. Compare values only for comparable test data, error assumptions, and units; absolute values are not model-quality grades.

These pooled supplied-error scores require a finite score from every requested split. They are unavailable if CV is disabled or the requested splits are incomplete. Ordinary CV means can still summarize finite split scores, so inspect the successful-split and finite-score counts before interpreting them. Supplied-error CV information criteria are not calculated; full-fit AIC, $\mathrm{AIC_c}$, and BIC remain available.

Metrics excluded from cross-validation comparisons

The app does not give fit/CV comparisons for adjusted R^2, information criteria (AIC, $\mathrm{AIC_c}$, and BIC), or either NRMSE normalization. These quantities depend on the sample size or subset-dependent reference values:

- **Adjusted R^2 and information criteria** incorporate the number of observations or fitted parameters as part of their calculation. Because the full-data model and each cross-validation model are fitted to different subsets of the data, these quantities are not on the same scale and are not directly comparable.
- **NRMSE** is normalized by a factor that generally differs across subsets. This means the full-fit and CV values can have different denominators, making a direct comparison misleading.

Eligible cross-validation metrics summarize prediction behavior during CV, while the ineligible CV metrics remain available for evaluating and comparing fitted models. Not all eligible metrics are available for every cross-validation scheme because some require more than one observation to be defined or stable. For example, R^2 cannot be calculated for a single omitted point and will be unavailable under leave-one-timepoint-out CV; interpretation then depends on the metrics that are reported.

A.7 Interpreting residual diagnostics

Residuals are observed minus fitted outputs. They use analyzed in vivo time units for Approach 1 and analyzed response units for Approach 2. The app provides raw residual and normal quantile-quantile (Q-Q) plots to explore patterns [27]. With few observations, random variation can resemble structure, so these are exploratory aids rather than pass/fail tests.

Residual-vs-time plots

Raw residuals retain the units of the analyzed fitted output. Each plot has its own symmetric axis and a minimum display range so tiny numerical errors are not magnified. Use the axes and ordinary error metrics when comparing magnitudes across fits. With unequal supplied SD/SEM, unequal raw residual spread can be consistent with the entered error model; raw residuals are not divided by those errors.

Q-Q plots

The Q-Q display centers residuals and divides them by their sample standard deviation before comparison with a normal reference. It does not standardize each residual by its supplied SD/SEM. With unequal measurement errors, the pooled residual distribution need not look normal even when individual errors follow the supplied normal model. Curvature, tail departures, or isolated points prompt review; a straight pattern does not establish independence, model correctness, or agreement with the supplied error scale. If residual variation is too small, the app reports that the display is not informative.

What to look for

Residual diagnostics are intended for qualitative review—by eye—rather than use as formal statistical tests. Quantitative tests are available for problems such as non-random errors, correlated errors across time (autocorrelation), and variance that changes across the response range (heteroscedasticity) [27]. However, with the small numbers of observations typical of IVIVC datasets, these tests may have limited ability to distinguish systematic problems from random variation. Visual patterns are prompts for further review rather than definitive findings. Residuals scattered around zero in the residual-vs-time plot, and points near the diagonal in the Q-Q plot, are consistent with a model form that captures the observed pattern, though they do not establish model validity. The following patterns may indicate bias or a model form that does not fully capture the data:

- Sustained curvature or systematic trends in the residuals over time.
- Clustering, where residuals at adjacent time points are consistently on the same side of zero.

When any of these patterns appear, they can motivate evaluation of whether a different model form or a transformed (i.e., normalized or scaled) response better describes the underlying relationship. For a visual example and additional references for interpreting residual plots, see Appendix B.

A.8 Interpreting parameter uncertainty

Model parameters cannot always be estimated precisely. Where available, the app reports each estimate $\pm$ the half-width of an approximate 95% confidence interval. The parameter standard error is the square root of its covariance diagonal entry; it is not the SD/SEM supplied for an observation. A wide coefficient interval indicates imprecision under the fitted model and error assumptions. A narrow interval is conditional on those assumptions and does not establish model correctness.

With supplied absolute errors, the coefficient interval uses a normal multiplier (approximately 1.96 at 95%), conditional on those errors. Without supplied errors, the app uses a Student t multiplier based on residual degrees of freedom $n - k$ when available, with its normal fallback otherwise. Replicate counts do not supply extra degrees of freedom or choose a different multiplier.

The covariance is a local approximation around the fitted parameters; nonlinear curvature or weakly determined parameters can make that approximation inaccurate [19]. Intervals can then understate asymmetry or dependence. Review them alongside parameter coupling, the fitted curves, and scientific plausibility [28, 29].

The app summarizes parameter uncertainty using informational badges as screening aids, not acceptance criteria. They highlight fitted-parameter uncertainty that may affect interpretation or prediction.

Diagnostic	What it summarizes	Interpretation guidance
Relative standard error	Parameter standard error expressed as a percentage of the absolute parameter estimate.	A large value means the uncertainty is large relative to the estimate, so the data are not pinning down that parameter well. The app flags values of at least 50% as elevated and at least 100% as large.
Parameter coupling / correlation	How strongly pairs of parameter estimates move together during fitting (pairwise correlation).	When two parameters are highly correlated, the model can reach a similar fit by trading off one against the other, making both harder to estimate reliably. Some coupling is normal in nonlinear models. The app reports correlation alone as *parameter coupling noted*; when high correlation coincides with elevated relative standard error or a numerical sensitivity flag, it reports a stronger identifiability caution. Absolute correlations of at least 0.90 are noted, with stronger concern when correlations of at least 0.95 occur alongside other uncertainty diagnostics.

If no parameter uncertainty concerns are flagged, the app did not detect large relative standard errors or strong parameter coupling under these screening checks. This does not establish mechanistic correctness, predictive adequacy for a particular use, or regulatory acceptability. Conversely, a coupling or warning badge does not by itself determine whether the model can support a particular use; rather, it identifies parameter behavior that can be reviewed alongside the fitted curves, residuals, cross-validation behavior, and scientific plausibility.

Parameter uncertainty deserves particular attention when a model will be interpreted mechanistically, used to extrapolate beyond the observed data range, or used to compare degradation behavior across materials, processes, or test conditions. In these situations, relevant considerations include whether the reported intervals are wide, parameters are strongly coupled, fitted parameter values are extreme or scientifically implausible, or small changes in the data or fitting settings produce substantially different estimates. Strong fit metrics alone may not be sufficient to address concerns about poorly estimated parameters or scientifically implausible values; these aspects warrant separate review.

A.9 How curve and prediction intervals are calculated

For fitted curves and parametric predictions, the app samples complete parameter vectors from a multivariate normal covariance approximation. This preserves estimated parameter relationships within a fit and dependence across times along each sampled curve. Separate calibration fits are sampled independently. When prediction SD/SEM is supplied, independent normal response errors are propagated once per entered row; inverse mapping uses the same sampled response to calculate its time and response coordinate.

Sampled bounds enclose the middle 95% of calculated values at each position. They are marginal, pointwise intervals, not a simultaneous whole-curve band or joint time-response region. Without future variability, time-constant and inverse response-axis bounds come directly from the supplied normal input error model. Selecting future variability adds independent normal future measurement noise to the propagated response draws; inverse future-response bounds use the same successful mappings as the time bounds. See Section 5.3 for the display legend and method-specific interpretation.

The current browser and CLI conventions use seed 12345 for these calculations, separately from CV and initialization seeds:

Calculation	Requested draws
Fitted-curve bands	1,000 per fit
Approach 2 response-value ratio intervals	1,000 per calibration fit
Time-constant table standard uncertainty	50 per fit
Approach 2 time-constant ratio intervals	1,000 per calibration fit
Approach 2 observed-response inverse intervals	100 from the in vivo fit

Central estimates use the fitted parameters and analyzed inputs, not averages across draws. Other than inverse time intervals and their optional future-response intervals, sampled intervals require every requested calculation to succeed at the affected position. Inverse time intervals can exclude unsuccessful mappings when the configured success threshold is met; these intervals are labeled conditional and describe successful mappings only. A valid center can remain without its interval. Sampling settings and failure details are retained with prediction results.

The integral time constant uses numerical integration from zero to infinity with the integration routine's default tolerances and a subdivision limit of 100. Warnings or invalid results make the time constant unavailable without discarding the fitted curve. Any invalid sampled time constant makes its table standard uncertainty unavailable, while retaining a valid central estimate. Time-ratio plot bounds separately propagate both parameter vectors through integration and their ratio; they do not combine the table SDs as independent errors.

Inverse mapping starts in the calibration time span and can extend it by doubling, with a current default of up to three extensions. Unscaled time has a lower search bound of zero, or the observed minimum if already negative; log-time coordinates have no zero-coordinate floor. The limit used is recorded with the result. The finite search and numerical monotonicity checks examine a branch of the fitted curve; they do not prove global uniqueness or that no solution exists outside the examined range.

A.10 Considering model complexity

More complex models (i.e., those with more parameters) may fit observed data more closely, but with small datasets this may be a liability rather than an asset. A flexible model may trace the noise in the data rather than the underlying pattern, and its parameters may be poorly estimated as a result, an issue called *overfitting*. Adjusted R^2, AIC, AIC_c, BIC, cross-validation metrics, and residual diagnostics provide complementary information about whether added complexity is justified [30].

When two models have similar fit and similar empirical support, model simplicity and scientific plausibility are among the factors that may support the rationale for the selected model.

A.11 Scientific plausibility and intended use

Statistical performance alone is not sufficient to evaluate a model. Scientific plausibility also depends on what is known about the material, degradation mechanism, and test conditions. A model that is statistically competitive but produces a fitted shape, parameter values, or extrapolated behavior that conflicts with known degradation behavior may warrant additional rationale regardless of its metrics.

The intended use of the model—including whether it will inform a regulatory submission—affects the type and depth of supporting evidence that may be relevant. Applicable considerations are described in FDA guidance on computational modeling credibility and reporting [1, 2].

A.12 Example model review workflow

The following is one possible sequence of review steps for illustrative reference. It does not establish a required procedure or replace applicable credibility and submission guidance [1, 2].

1. Confirm that the dataset, preprocessing choices, and selected response values (the measured values being modeled) are appropriate for the intended analysis.
2. Review fitted curves and residual plots for each candidate model, checking for systematic patterns or implausible fitted shapes.

3. Compare goodness-of-fit metrics such as R^2 and RMSE in the context of assay variability, endpoint importance, and intended use.
4. Review cross-validation performance and fit/CV ratios to assess sensitivity to the available data.
5. Review relative model evidence using information criteria to compare fit quality against model complexity.
6. Review fitted-parameter estimates, 95% confidence intervals, uncertainty diagnostics, and scientific plausibility.
7. Consider whether model complexity is justified by the available data and the degradation mechanism.
8. Document the rationale for the user-selected model based on a balance of statistical performance, scientific plausibility, and intended use.

A.13 Questions to guide model review

The following questions illustrate considerations that a user might explore. They do not define a required review process or imply that each question must be addressed in a regulatory submission.

1. **Does the model describe the observed data well?**
   - Does the fitted curve follow the main trend in the data?
   - How do performance metrics such as R^2, RMSE, and NRMSE compare in the context of the intended use?
   - How large is the prediction error relative to assay variability, measurement uncertainty, or the degradation endpoint being evaluated?
   - Are there regions where the model consistently overpredicts or underpredicts?
2. **Does the model generalize beyond the fitted data?**
   - Are cross-validation metrics similar to the metrics for the model fitted to the full dataset, and are fit/CV ratios near 1?
   - Could the cross-validation result be sensitive to the splitting scheme, number of observations, or a small number of influential points?
   - If cross-validation performance is poor, how does that affect interpretation of the model for its intended purpose, including whether a simpler or more stable form may warrant consideration?
3. **Are the model parameters interpretable and stable?**
   - Are the fitted parameter values scientifically plausible?
   - What do the 95% confidence intervals indicate about the precision of the parameter estimates?
   - Are any parameters strongly coupled—meaning different combinations of parameter values could produce a similar fitted curve?
   - If coupling is present, does it coincide with large parameter uncertainty or an identifiability caution?
   - Are individual parameter values being interpreted mechanistically, or is the model being used mainly to describe the overall curve shape?

   A model can have a fitted curve that closely follows the observed data while still having uncertain or coupled parameters. In that case, the curve may be useful for empirical description within the observed data range. Interpret individual parameter values cautiously.
4. **Is the model unnecessarily complex?**
   - Does the more complex model improve fit, cross-validation behavior, or relative model evidence enough to support the added parameters?
   - Do AIC, AIC_c, or BIC provide support for the additional complexity within the candidate model set?
   - If two models perform similarly, how do model simplicity and scientific plausibility contribute to the rationale for the selected model?

- Does the dataset contain enough observations to inform the fitted parameters?

5. **Is the model scientifically plausible?**
   - Is the fitted curve shape consistent with expected material degradation behavior?
   - How do the model assumptions align with the material, test conditions, and degradation endpoint?
   - How does the model behave near the edges of the observed data range, and how would that behavior affect interpolation or extrapolation?

   A statistically competitive model may still warrant additional rationale if its shape, parameter values, or extrapolated behavior conflict with the known degradation process.

6. **How does the intended use affect interpretation of the model?**
   - Is the model being used for exploratory analysis, interpolation, prediction, comparison across conditions, or support for a regulatory rationale?
   - How does the level of uncertainty affect interpretation for that purpose?
   - Are the model's limitations clearly documented?
   - Would an independent reviewer understand the rationale for selecting this model over other candidate models?

Potential elements of a model-selection rationale include fitted performance, cross-validation behavior, relative empirical support, parameter diagnostics, residual behavior, scientific plausibility, and intended use.

A.14 Example interpretation statements

The following passages illustrate how model results might be described in a technical context. They are examples only and are not FDA-endorsed language for regulatory submissions. Adapt all language to the specific analysis and supporting evidence.

- Model A had the lowest RMSE and highest adjusted R^2 among the evaluated models, indicating the strongest goodness-of-fit under the selected preprocessing conditions.
- Model A was selected over the more complex Model B because both models showed similar fit and AIC_c support (Delta $AIC_c < 2$), so the observed data provided little additional support for the added complexity of Model B.
- Although Model C had strong goodness-of-fit, its cross-validation error was substantially higher than its fitted error, suggesting sensitivity to which observations were included in fitting.
- Residuals for Model A were scattered around zero without obvious systematic patterns, consistent with a model form that captures the observed data.
- Model B showed a systematic trend in the residuals across time, suggesting the selected model form does not fully capture the degradation profile.
- The fitted parameters for Model A were within plausible ranges for the degradation mechanism, with narrow local confidence intervals under the selected error assumptions.
- Model C showed strong parameter coupling between the rate and shape parameters, with relative standard errors exceeding 100%. Parameter estimates were retained for curve description but were not interpreted individually.
- The selected model was chosen based on a balance of goodness-of-fit, cross-validation behavior, relative empirical support, parameter stability, and scientific plausibility for the degradation mechanism and intended use.

B Worked Example: Comparing Candidate Models

B.1 Purpose of the example

This worked example illustrates how fitted curves, performance metrics, cross-validation, residual plots, Q–Q plots, and parameter uncertainty results can be considered together when comparing candidate models. It provides practical examples of residual patterns that can help readers interpret their own results; further illustrated examples are available in [28, 27, 31, 32, 33]. The dataset is simulated so the underlying relationship is known, but the comparison is presented in the same way as an analysis of an unknown experimental relationship. The example is intended as a practical demonstration, not as an acceptance procedure or a set of model-selection criteria.

An accompanying Python script generates a noisy single-exponential decay dataset and fits four candidate models. The four candidate models fitted to the primary dataset are:

- a linear model, included as a form that does not match the data-generating process and is expected to demonstrate clear lack of fit;
- an exponential model, which matches the form used to generate the data;
- a stretched-exponential model, which can reduce error slightly by adding a shape parameter; and
- a double-exponential model, which can reduce error further while adding parameters that may be weakly estimated and strongly coupled.

This example uses the same fitting functions, performance metrics, shuffle-split cross-validation constructor, and relative evidence calculations used by the app, without supplied SD/SEM. The figures use a simplified layout rather than reproducing the interactive app displays exactly.

B.2 Simulated dataset and analysis settings

The simulated dataset has the form

$$y = a \exp(bt) + \varepsilon,$$

where ε is independent Gaussian noise. The random seed, parameter values, noise level, and cross-validation settings are fixed so the example is reproducible.

Setting	Value
Generating model	Exponential
True parameter a	100.000
True parameter b	-0.075
Noise standard deviation	3.000
Dataset random seed	27
CV scheme	Shuffle split
CV splits	100
CV test fraction	0.25
CV random seed	12345

B.3 Fitted curves

Figure 1 shows the simulated observations with all four fitted curves. The fitted curve display is the first opportunity to identify clearly implausible behavior, but models that differ meaningfully can appear similar when the response scale is large relative to the residual differences. For this example, the linear model visibly misses the decay shape. The three exponential-family models can appear similar at the scale of the response plot, even though the stretched and double-exponential forms use additional parameters.

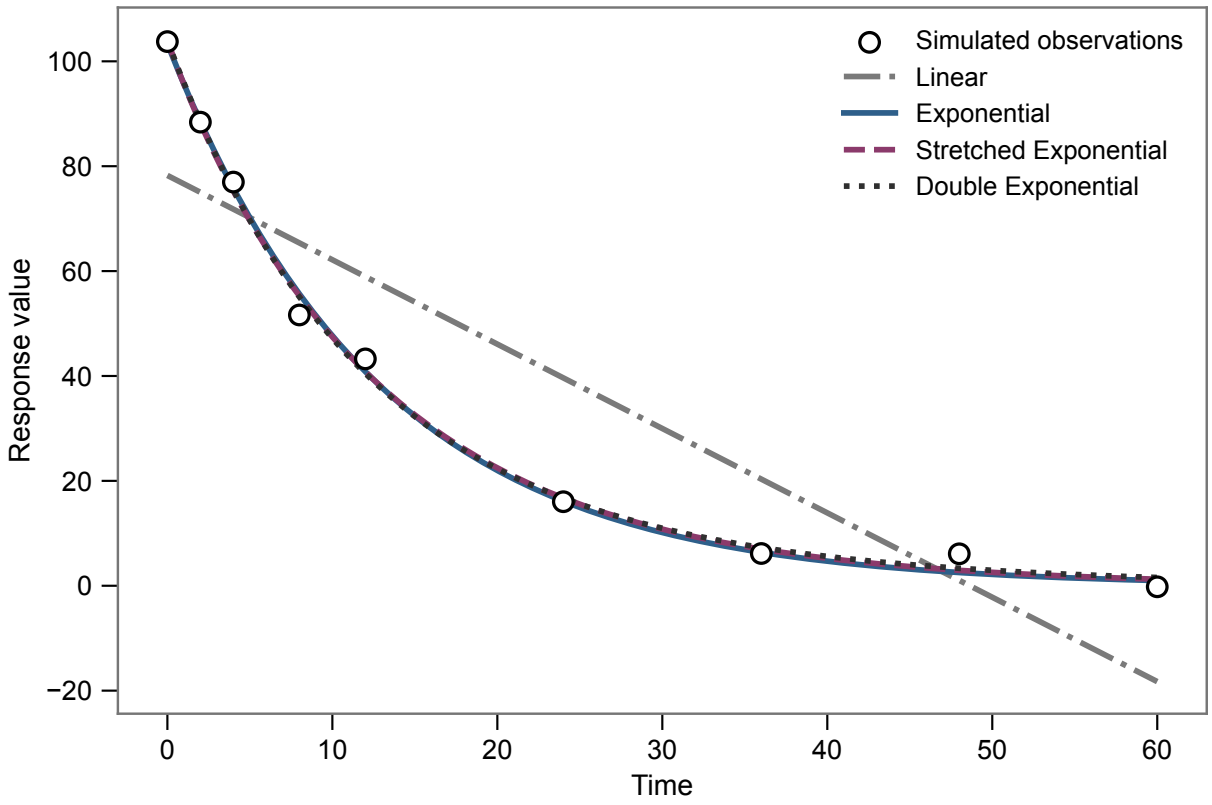


Figure 1: Candidate models fitted to the simulated dataset.

B.4 Fit metrics, model complexity, and cross-validation

Table 22 reports goodness-of-fit metrics, AIC_c comparisons, cross-validation error, and a compact parameter-correlation summary. These quantities answer different questions. R^2 and RMSE describe agreement with the fitted observations. AIC_c compares relative empirical support while penalizing additional fitted parameters. Cross-validation RMSE describes prediction error for omitted observations. The ratio of cross-validation RMSE to full-fit RMSE provides a simple indication of how much prediction error increases when part of the dataset is excluded from fitting.

Table 22: Model performance summary.

Model	R^2	RMSE	AIC_c	Delta AIC_c	CV RMSE	CV/fit
Linear	0.805	16.338	56.283	37.381	24.279	1.486
Exponential	0.997	2.048	18.902	0.000	2.783	1.359
Stretched Exponential	0.997	1.987	23.156	4.254	3.557	1.791
Double Exponential	0.997	1.960	30.112	11.209	1261.292	643.566

In addition to its visible lack of fit, the linear model has substantially poorer fit metrics. Its R^2 remains moderately high because it captures part of the large overall change in response, even though it does not describe the curved decay pattern well. The exponential model provides the strongest overall balance: it represents the generating relationship with only two parameters and has the lowest AIC_c. The stretched and double-exponential models reduce RMSE slightly, but their higher AIC_c values indicate that these small improvements do not offset the added complexity. Cross-validation provides a separate check on how prediction error changes when observations are omitted from fitting. The cross-validation results are similar to the full-fit results for the linear, exponential, and stretched-exponential models. In contrast, the double-exponential model gives much poorer cross-validation performance, indicating greater sensitivity to the observations used for fitting.

B.5 Residuals versus time and missed structure

Residuals are calculated as observed minus predicted response (i.e., the prediction errors). A fitted form that captures the observed pattern generally leaves residuals distributed around zero without sustained curvature or adjacent groups that remain mainly above or below zero. Figure 2 uses the same y-axis range for all four models so the sizes and patterns can be compared directly.

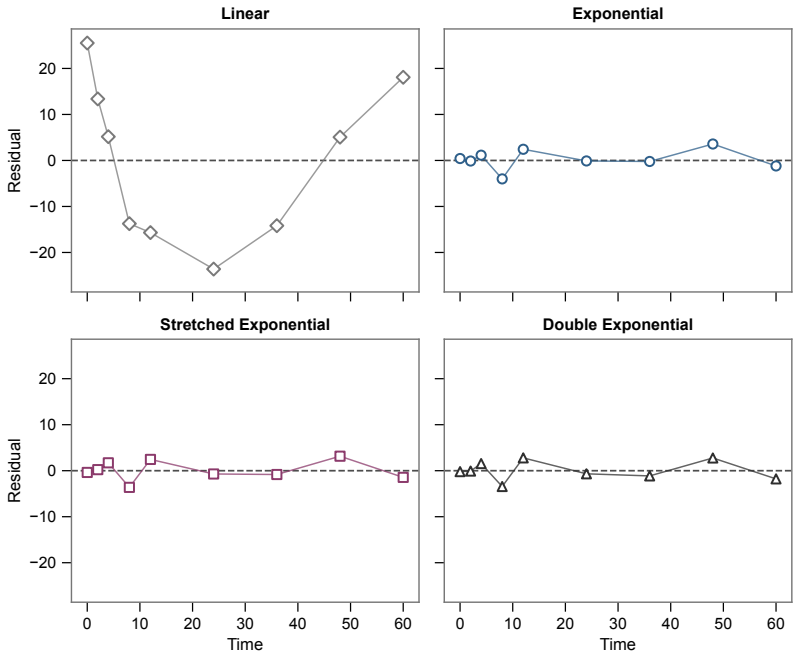


Figure 2: Residuals versus time for the four fitted models.

The linear residuals show a pronounced systematic pattern, indicating that the model does not represent the curved decay well. The exponential residuals fluctuate around zero without any sustained structure. The stretched and double-exponential models produce similarly centered residuals that are slightly smaller at some observations; that modest improvement can be interpreted alongside their added complexity, AIC_c, cross-validation behavior, and parameter uncertainty.

B.6 Q–Q plots and residual distribution shape

A normal Q–Q plot compares ordered standardized residuals with the quantiles expected from a normal distribution. Points close to the reference line are broadly consistent with the assumed residual distribution. Curvature, separation at the ends, or an isolated point can draw attention to skewness, heavy tails, or an unusual residual. With a small dataset, the plot has limited resolution and is interpreted together with the residual-versus-time plot rather than as a formal confirmation of normality.

These Q–Q plots do not distinguish the candidate models as clearly as the residual-versus-time plots, largely due to the small number of observations. A Q–Q plot describes the residual distribution, whereas a residual-versus-time plot is more directly informative about systematic lack of fit across the observed profile. For this dataset, the app would not display Q–Q plots for the exponential-family models because the residual variation is too small relative to the overall response range. Instead, it would report that the Q–Q diagnostic is not informative rather than magnifying very small numerical differences.

B.7 Effect of the number of observations

Figures 1–3 use the primary dataset with 9 observations. To illustrate the effect of denser sampling, a second independent dataset was generated from the same single-exponential relationship and noise distribution at 61 evenly spaced times from 0 to 60. Figure 4 shows the fitted curves and residual diagnostics for the linear and exponential

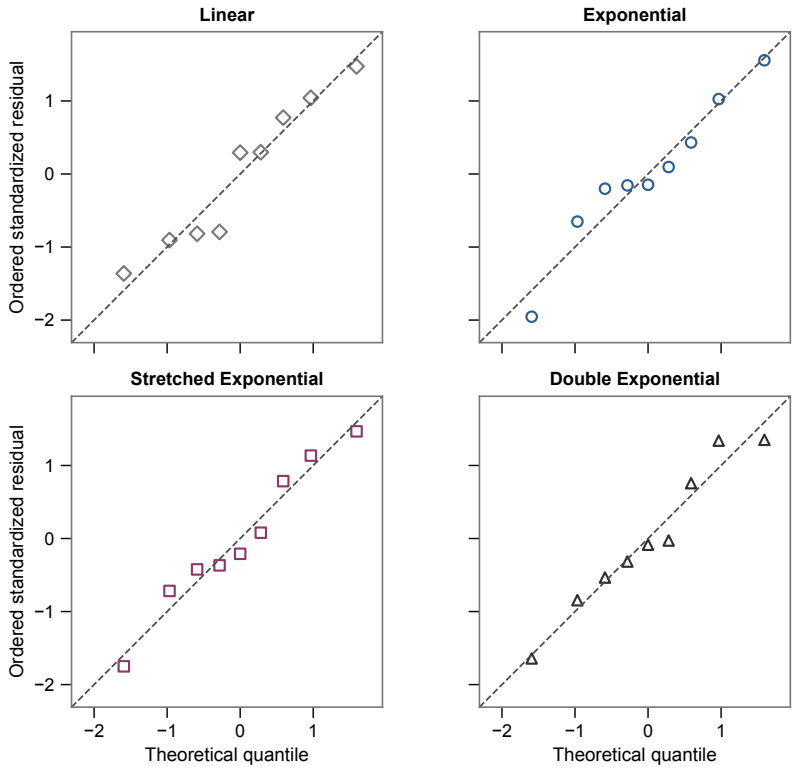


Figure 3: Normal Q–Q plots of standardized residuals.

models. This section provides a qualitative example of how the number of observations can affect the ability to distinguish a model that captures the observed pattern from one with systematic lack of fit.

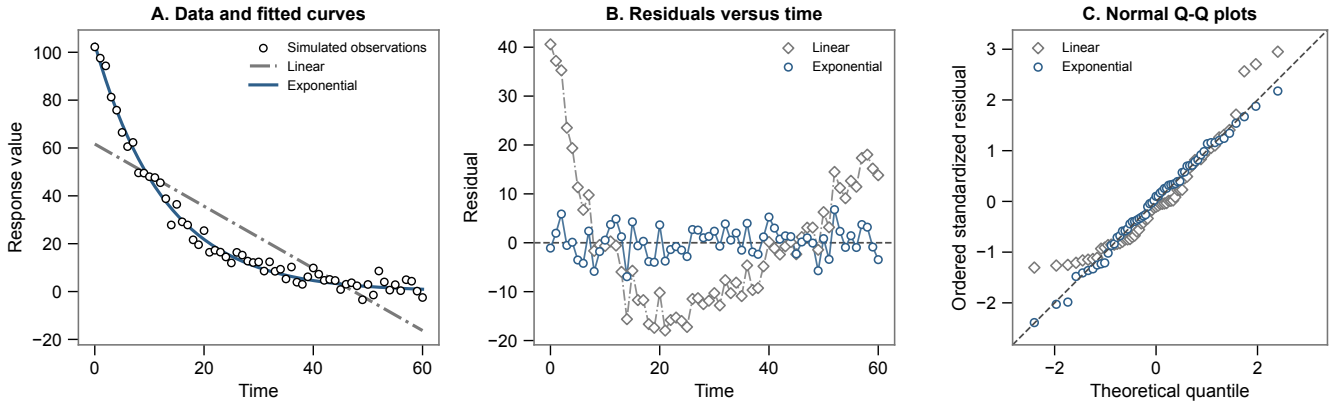


Figure 4: Graphical diagnostics for linear and exponential fits to a simulated dataset with 61 evenly spaced observations: fitted curves, residuals versus time, and normal Q–Q plots.

The smaller dataset already suggests that the linear model does not capture the decay relationship, but the denser dataset makes the lack of fit easier to diagnose. With more observations, the linear residuals form a clearer systematic pattern over time rather than fluctuating randomly around zero, and departures from the Q–Q reference line are easier to assess. In contrast, the exponential-model residuals remain comparatively patternless around zero and follow the Q–Q reference line more closely. Additional observations do not guarantee that a selected model captures the underlying relationship, but they provide more information for distinguishing systematic lack of fit from random variation. With sparse data, the same patterns may be less apparent or more easily mistaken for noise.

B.8 Fitted-parameter uncertainty

The app reports each fitted parameter as the estimate ± the half-width of an approximate 95% confidence interval. Table 23 lists that half-width separately from the estimate, together with the standard error (SE) and relative standard error (RSE = SE / absolute estimate in %). These intervals describe individual parameters, not the scatter of future observations. This tutorial leaves future measurement variability unselected. The app's fitted-curve bands and default prediction error bars propagate parameter uncertainty through the calculation (Section 5.3); this tutorial's curve figures show fitted lines without uncertainty bands.

Table 23: Fitted-parameter estimates and approximate uncertainty.

Model	Parameter	Estimate	SE	95% CI half-width	RSE (%)
Linear	a	78.242	8.936	21.131	11.421
Linear	b	-1.609	0.300	0.709	18.628
Exponential	a	103.344	1.758	4.157	1.701
Exponential	b	-0.077	0.003	0.008	4.257
Stretched Exponential	a	104.141	2.281	5.580	2.190
Stretched Exponential	b	-0.085	0.014	0.034	16.204
Stretched Exponential	c	0.964	0.057	0.139	5.874
Double Exponential	a	64.400	355.320	913.379	551.739
Double Exponential	b	-0.096	0.156	0.401	161.667
Double Exponential	c	39.562	356.112	915.415	900.144
Double Exponential	d	-0.056	0.133	0.343	238.297

Table 24 reports the strongest pairwise parameter correlation for each fitted model. Strong correlation can indicate that different combinations of parameter values produce similar fitted curves, especially in flexible nonlinear models. The app considers correlation together with relative standard error: high correlation alone is not unusual, but high correlation combined with large parameter uncertainty provides stronger evidence that the model parameters cannot be uniquely identified.

Table 24: Largest absolute pairwise parameter correlation for each fitted model.

Model	Maximum absolute correlation	Parameter pair
Linear	0.723	a-b
Exponential	0.604	a-b
Stretched Exponential	0.963	b-c
Double Exponential	1.000	a-c

The exponential parameters are expected to be estimated comparatively precisely because the fitted model matches the generating relationship, and indeed their relative standard errors are only a few percent. For the stretched exponential, the estimated shape parameter remains close to the single-exponential value of 1, so the added parameter changes the fitted curve only slightly and does not improve AIC_c. In the double-exponential fit, both components retain nonzero amplitudes (a and c), but the amplitudes and rate parameters (b and d) have very large uncertainty and are almost perfectly coupled. The fitting procedure can trade one component against the other while producing nearly the same overall curve. In the app, this combination of high parameter correlation and elevated relative standard errors would receive an *identifiability caution* badge. This illustrates why a slightly closer fitted curve does not necessarily support the added model structure or interpretation of the individual component parameters.

B.9 Integrated interpretation

No single output determines the preferred candidate. In this example, the linear model is readily set aside because the fitted curve and residual pattern show clear lack of fit. The exponential model provides the strongest overall balance: it captures the curve shape without obvious residual structure and does so with the fewest parameters

among the plausible candidates. The stretched and double-exponential models can achieve slightly lower error, but their higher AIC_c values indicate that the improvement does not justify the added complexity for this dataset. The double-exponential parameter results provide an additional caution because both fitted components remain active while their amplitudes and rates are extremely uncertain and strongly coupled.

For an experimental analysis, the same categories of information can be considered without knowing the generating model: fitted curves, complementary metrics, cross-validation as a sensitivity diagnostic, residual-versus-time and Q–Q plots, parameter uncertainty, and scientific plausibility can all contribute to the rationale for a user-selected model.

C Worked Example: Sampling Times, Noise, and Model Uncertainty

C.1 Purpose and setup

How well can a fitted curve recover behavior between observations? This example uses a known synthetic decay to explore how time-point placement, measurement noise, and replicate count affect that recovery. It complements model comparison: even when the correct curve form is available, the observations may provide too little information to determine its shape reliably.

This repeated-study illustration is separate from the app's optional SD/SEM input and propagated-interval calculations.

The synthetic fraction remaining follows the normalized logistic model,

$$F(t) = \frac{1}{1 + \exp((t - t_{50})/s)}, \qquad t_{50} = 50, \quad s = 8.$$

Here, t_{50} is the time at which half remains, and s controls the transition width. The upper and lower asymptotes are fixed at 1 and 0; t_{50} and s are fitted using the app's model-fitting machinery. Time is in illustrative units.

Each sampling condition was simulated 100 times with independent Gaussian measurement errors and fitted with the same logistic model. Standardized errors were shared across conditions to aid comparison. A separate simulated study supplies the illustrative observations. Each replicate is an independent measurement at its assigned time; noise also applies at time zero. The response scale is fixed, rather than obtained by dividing by a noisy initial observation, and noisy observations are not clipped to the range 0–1.

Table 25: Sampling conditions. Panel letters refer to Figure 5. Noise SD is on the fraction-remaining scale.

Panels	Condition	Times	Replicates/time	Noise SD
A, B	Reference	0, 20, 40, 100	1	0.03
C	Misplaced focus	0, 10, 20, 100	1	0.03
E	Transition focus	0, 40, 60, 100	1	0.03
G	More time points	0, 20, 40, 60, 80, 100	1	0.03
D	Higher noise	0, 20, 40, 100	1	0.09
F	More replicates, higher noise	0, 20, 40, 100	3	0.09
H	More times, higher noise	0, 20, 40, 60, 80, 100	1	0.09

The reference is a deliberately simple, sparse schedule, resembling an in vivo time course with several early observations and fewer later observations. Its times are illustrative, rather than a recommended protocol.

C.2 What the fitted curves show

A convincing individual fit can hide weak support between observations. The reference fit in Figure 5A follows the observed points closely. Across repeated studies, however, the fitted transition varies substantially (B). Moving the early observations farther from the transition ("misplaced focus," C) leaves even greater freedom in when and how rapidly the decline occurs. Agreement with the available observations does not establish the behavior within the long unsampled interval.

Moving a time point can be as useful as adding time points. Replacing time 20 with time 60 places observations on both sides of the main transition (E). Recovery improves markedly with the same number of observations. Adding times 60 and 80 to the reference also improves recovery (G). Both schedules track the true curve closely, with much less variation between studies. Their success reflects coverage of the changing part of the curve, rather than simply a larger sample count.

Replicates help with noise, but leave gaps in time coverage. Higher measurement noise worsens recovery under the reference schedule (D). Additional replicates at those same times improve recovery somewhat (F), but substantial uncertainty about the transition remains. Adding informative times instead gives more consistent recovery (H), despite using fewer total observations than the triplicate schedule. Better coverage also partly offsets the increased noise when compared with the original reference (B). These results show the importance of placement in this example; they do not imply that replication or measurement precision is generally unimportant.

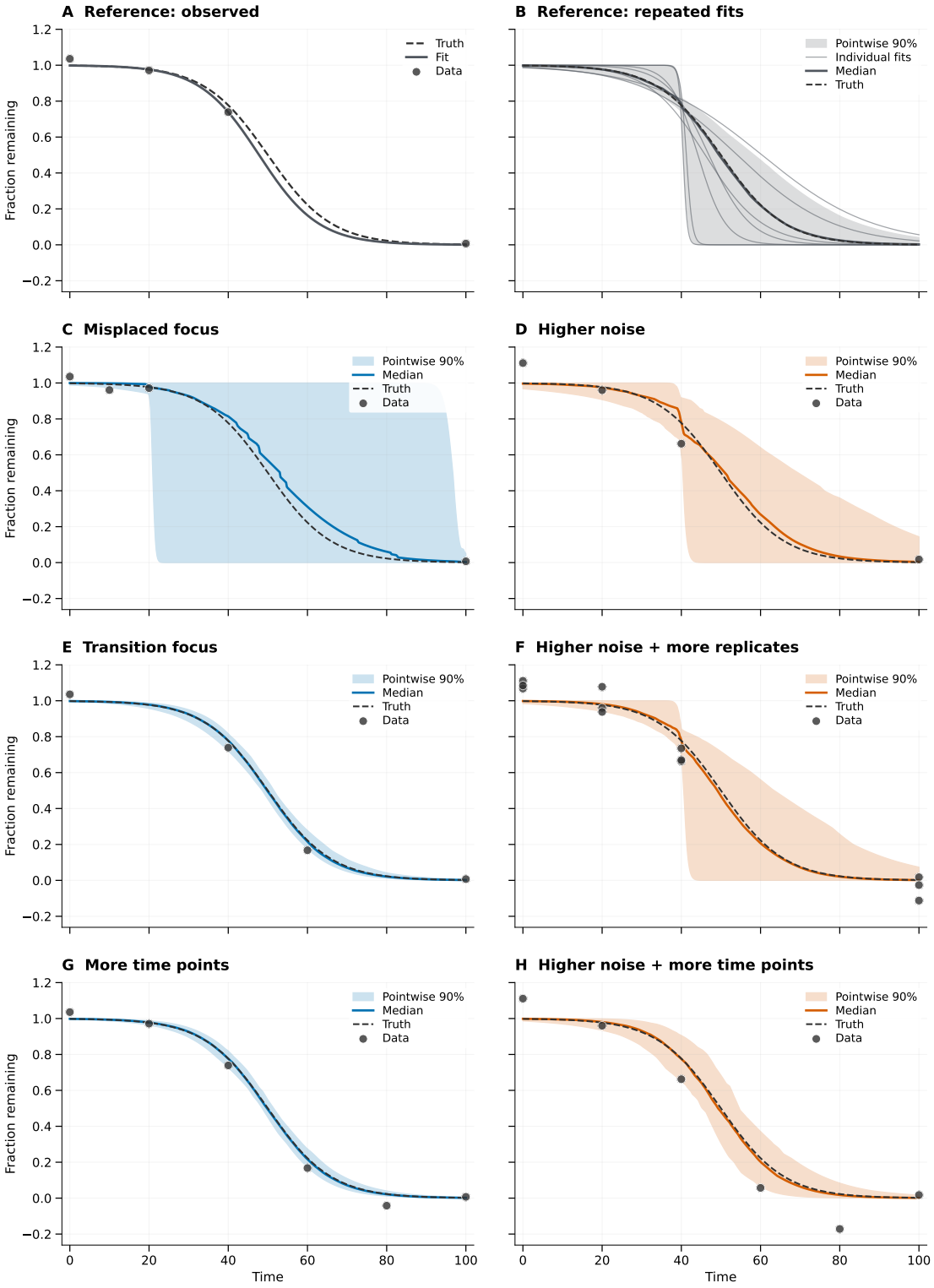


Figure 5: Recovery of the known logistic decay under different sampling conditions. A shows one study and its fitted curve. In B–H, the line and shaded area show the pointwise median and middle 90% of fitted curves across 100 repeated studies. Points in C–H come from the separate illustrative study. B also shows selected individual fits spanning the observed shape variation; these are not frequency-weighted examples. The shaded areas describe variation between simulated studies, not the app's 95% parameter-uncertainty bands, observation prediction intervals, or cross-validation variability from one dataset.

C.3 Recognizing weak support in the app

Figure 6 summarizes the approximate 95% confidence-interval half-width for t_{50}, the same type of interval shown in the app's parameter tables. It describes uncertainty in this one fitted parameter, rather than the shaded curve intervals shown in the app. The reference and misplaced-focus schedules often give wide or unavailable intervals. Transition-focused and denser sampling give much narrower intervals; denser sampling generally narrows them further. Under higher noise, adding times gives more consistently narrow intervals than adding replicates at the reference times. The latter still produces some extremely wide intervals, even though a numerical interval can be calculated.

These are local approximations based on fitted-parameter covariance, and can be unreliable with sparse data. The boxplots summarize only available intervals; unavailable intervals in the reference, misplaced-focus, and higher-noise cases can also signal weak support. A narrow interval alone does not establish that the curve form is appropriate.

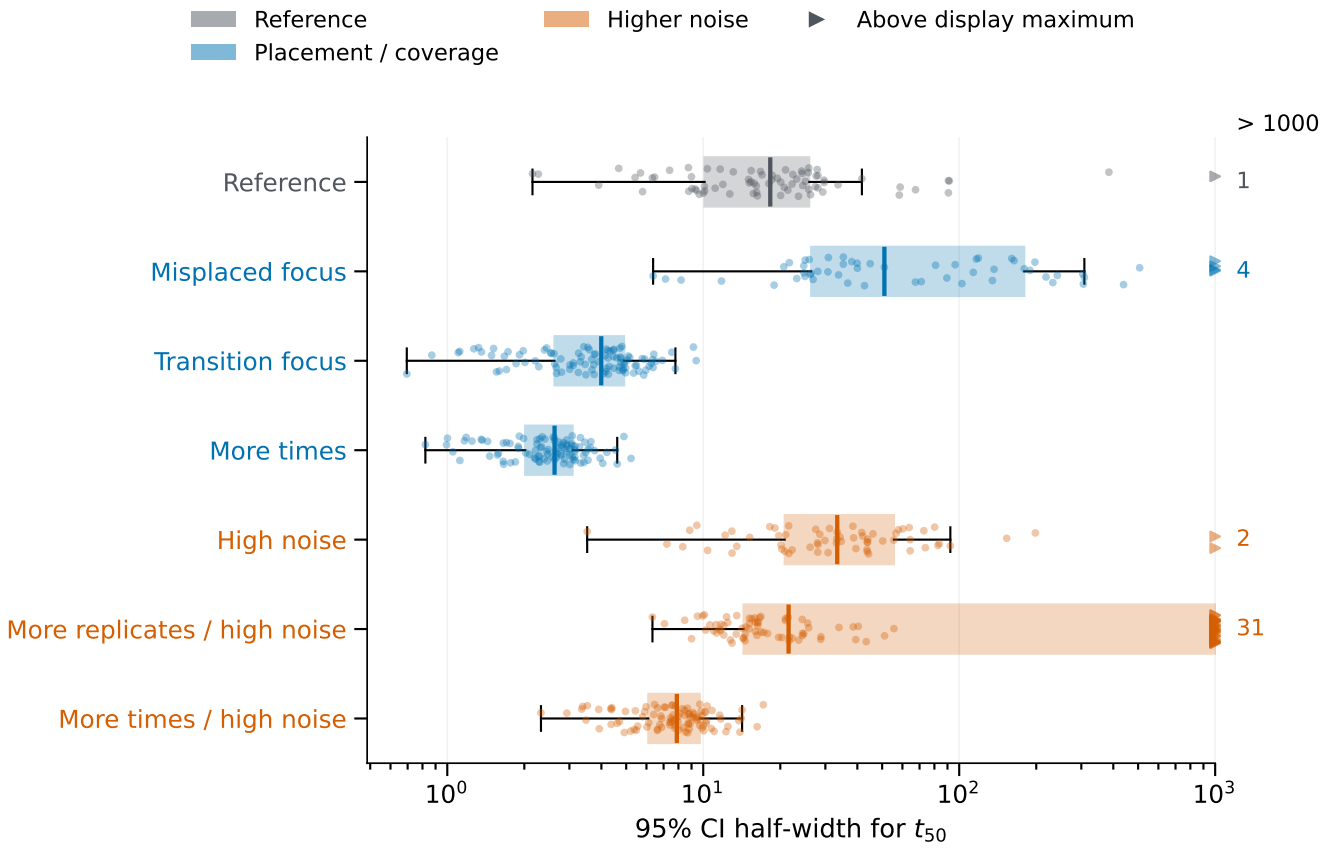


Figure 6: Uncertainty in the fitted transition time. Each point represents an available local 95% confidence-interval half-width for t_{50} from one repeated study; boxes summarize their distribution. Smaller values indicate more precise estimates under the fitted model. The horizontal axis is logarithmic. Right-edge triangles and counts mark values above the display limit; unavailable intervals are excluded.

For an actual dataset, inspect where observations fall relative to the fitted transition, the uncertainty in parameters describing its timing and width, and how the curve changes when observations or time points are omitted during cross-validation. Large parameter uncertainty or strong sensitivity to omissions can motivate additional sampling. Cross-validation cannot supply observations within an unsampled gap, and its variability is distinct from the repeated-study bands shown here.

The favorable transition-focused schedule uses knowledge of where the synthetic transition occurs. In practice, prior information, pilot observations, or broader initial coverage can help locate important changes. Comparing candidate schedules across several plausible transition times and curve shapes would provide a stronger basis for study planning. This example illustrates that reasoning for one known curve and error model; it does not identify a universally best schedule. The same sampling principles apply to other model forms, although useful sampling times and the size of the improvement will depend on the curve and noise.

D Glossary

The following definitions apply specifically within the IVIVC app and this user guide. They are intended as operational quick references for tool users and do not constitute official FDA regulatory definitions. See Appendix A for more detailed interpretation guidance.

Term	Meaning in this app and guide
Analysis approach	The overarching method used to establish or evaluate the in vitro/in vivo relationship. An approach can use fitted mathematical models or, for Approach 3, direct interpolation-based mappings.
Approach 1: Direct Correlation	Relates in vitro and in vivo times at common response values and fits a model that maps in vitro time to predicted in vivo time. Interpolation by response value can be used when exact response-value matches are unavailable.
Approach 2: Independent Model Fitting	Fits the same model family separately to the preprocessed observed in vitro and in vivo datasets. It supports supplied response SD/SEM for fitting and prediction. Interpolated points are not used for fitting.
Approach 3: Direct Mapping	Uses response-at-shared-time and time-at-shared-response-value mappings between the in vitro and in vivo datasets rather than fitting a parametric model.
Candidate model	A mathematical form considered for fitting within an applicable analysis approach. Candidate models can differ in shape, number of parameters, and flexibility.
Confidence interval (parameter)	Approximate interval for one fitted coefficient, reported as the estimate $\pm$ the half-width of a 95% interval when available. Supplied-error intervals use a normal multiplier conditional on the given SD/SEM; residual-based intervals use residual degrees of freedom when available.
Covariance matrix	Matrix summarizing approximate parameter variances and how parameter estimates vary together. With supplied errors, it uses the absolute SD/SEM scale; otherwise the error scale is estimated from residual variation. It supports parameter standard errors, intervals, and coupling diagnostics.
Cross-validation (CV)	Prediction-error and sensitivity diagnostic in which part of the fitting data is set aside, the model is fit to the remaining observations, and predictions are compared with the omitted observations. CV does not estimate parameter uncertainty.
Densified interpolation	Interpolation option that adds intermediate grid points in both the time and response-value directions before interpolation. The added values are estimates, not additional experimental observations.
Empirical support	Relative support from the observed data for a model balancing fit quality and complexity well within the displayed candidate set, under a selected information criterion. It does not establish correctness or suitability for a particular use.
Extrapolation	Use of a fitted relationship outside its observed calibration range. It relies on assumptions about behavior beyond the available data; a narrow interval does not establish that an extrapolated result is reliable.
Fit/CV ratio	Ratio of a full-fit metric value to the corresponding mean cross-validation metric reported by the app. Values near 1 indicate similar full-fit and cross-validation performance; interpretation also depends on the metric and CV scheme.
Fitted model	A candidate model after its parameters have been estimated from the data.
Fitted parameter diagnostics	App summaries that flag large relative standard errors, strong parameter coupling, or combined identifiability concerns for closer review. They do not determine whether a model is valid or acceptable.
Fitting dataset	Required dataset containing paired in vitro and in vivo time-course data used to establish the IVIVC relationship.
Full fit	Model fit obtained using the complete fitting dataset rather than a training subset created for cross-validation.

Term	Meaning in this app and guide
Goodness of fit	Degree to which a fitted model describes the observations used to estimate its parameters. The app summarizes goodness of fit with selected metrics and fitted-curve displays.
Grid search initialization	Procedure that evaluates many candidate parameter starting values before the final nonlinear least-squares fit and selects a promising starting point. It can improve fitting robustness but does not guarantee a global optimum.
Identifiability	Degree to which the available data distinguish among possible values of fitted model parameters. Poor identifiability occurs when substantially different parameter combinations can produce very similar fitted curves.
Information criterion	Model-comparison measure, such as AIC, AIC_c, or BIC, that combines fit quality with a penalty for model complexity. Information-criterion values are interpreted comparatively, not as absolute fit scores.
Interpolation	Estimation of response values between measured points. Interpolated values are derived from the observed data and are not new measurements.
In vitro	Measurements or behavior observed in a laboratory or other setting outside the living organism. In this app, in vitro data provide one side of the IVIVC relationship.
In vivo	Measurements or behavior observed in a living organism. In this app, in vivo data provide the other side of the IVIVC relationship.
IVIVC (in vitro–in vivo correlation)	Quantitative relationship between in vitro and in vivo degradation behavior. The app provides several approaches for establishing, evaluating, and applying such relationships.
Marginal interval	Interval describing one quantity separately. The displayed time and response bounds are marginal intervals; together they do not form a joint 95% region for both coordinates.
Measurement uncertainty (supplied)	Absolute SD for an individual response or SEM for a reported mean, entered as a constant or a column. Approach 2 treats these errors as given and assumes independent measurement errors across entered rows and between prediction and calibration data.
Model complexity	Flexibility introduced by a model's form and number of fitted parameters. Additional complexity can improve in-sample fit but can also increase parameter uncertainty or overfitting.
Model parameter	Numerical quantity in a model equation that is estimated from the data during fitting, such as an amplitude, rate constant, exponent, or lag parameter.
Normalization	Preprocessing that expresses responses relative to a reference, such as the first response or observed range. With supplied errors, the app rescales SD/SEM with the responses while treating the normalization reference values as fixed; uncertainty in those references is excluded.
Observation weight	Influence assigned to a fitting observation. With supplied SD/SEM σ_i, Approach 2 uses weights $1/\sigma_i^2$: smaller errors give more influence. These differ from information-criterion evidence weights.
Observed-response inverse mapping	Approach 2 method that estimates when the fitted in vivo curve reaches a supplied in vitro response, without using the prediction time or fitted in vitro curve. Horizontal bounds propagate in vivo parameter uncertainty and any supplied response uncertainty; vertical bounds describe the input response only. Unreliable or nonunique mapped times are unavailable.
Overfitting	Situation in which a model's flexibility allows it to follow random variation in the fitted data rather than only the underlying relationship. Overfitting can appear as excellent full-fit metrics accompanied by poorer CV performance or unstable parameters.
Parameter coupling / correlation	Tendency for two fitted parameter estimates to change together. Strong coupling can mean that one parameter can compensate for another while producing a similar fitted curve.

Term	Meaning in this app and guide
Parameter estimate	Fitted numerical value of a model parameter obtained from the optimization procedure.
Parameter uncertainty	Uncertainty in the numerical values of fitted model parameters. The app summarizes it with approximate confidence intervals, relative standard errors, and parameter-coupling diagnostics where available.
Parametric model	Mathematical model described by an equation with a finite set of fitted parameters. Approaches 1 and 2 can use parametric models; Approach 3 uses direct mappings rather than a fitted parametric model.
Pointwise interval	Interval applying at one plotted position. Approximate 95% pointwise bounds do not provide simultaneous 95% coverage of the whole fitted or predicted curve.
Prediction band	Interval intended to include the additional scatter of future observations. Approach 2 can include future measurement variability in pointwise response prediction intervals when selected; these do not form a simultaneous whole-curve band.
Prediction dataset	Optional in vitro dataset used after model fitting to generate corresponding in vivo predictions.
Prediction error	Difference between an observed output and its model prediction. The output may be time or response, depending on the approach and method. For observations used in fitting, this difference is also called a residual.
Preprocessing	Operations applied to uploaded data before fitting, comparison, or prediction. In this app, preprocessing can include normalization, scaling, and interpolation.
Propagated uncertainty	Uncertainty carried through a calculation from the relevant fitted parameters and, for Approach 2 when supplied, prediction-response SD/SEM. Its meaning depends on the prediction method and output coordinate.
Q–Q plot (quantile–quantile plot)	Graph comparing ordered residuals, centered and divided by their sample SD, with normal quantiles. It is a qualitative diagnostic, not a normality pass/fail test. The app does not scale these residuals by supplied observation SD/SEM.
Random seed	Starting value used by a pseudorandom procedure so randomized operations, like shuffle split CV and grid search, can be reproduced.
Relative model evidence	Comparison of candidate models using AIC, AIC_c, or BIC, including delta values, weights, and evidence ratios. These quantities describe relative support only within a comparable candidate set.
Relative standard error (RSE)	Parameter standard error expressed as a percentage of the absolute parameter estimate. Large RSE values indicate that uncertainty is large relative to the fitted value.
Replicate count	Optional positive integer recorded with a reported mean. Counts are metadata: they neither divide an entered SEM again nor add fitting observations.
Residual	Observed output minus the fitted output. In Approach 1 this is an in vivo time difference; in Approach 2 it is a response difference. Patterns can reveal systematic lack of fit that is not obvious from the fitted curve alone.
Residual degrees of freedom	Number of fitted observations minus the number of estimated curve parameters, $n - k$. It is used for residual variance and residual-based parameter intervals. With supplied absolute SD/SEM, coefficient intervals use a normal multiplier instead.
Residual plot	Plot of residuals against the fitting variable on the analyzed scale. Raw residuals use analyzed in vivo time units for Approach 1 and analyzed response units for Approach 2.
Response value	Measured y-axis quantity that changes with time, such as mass remaining, molecular weight, a mechanical property (e.g., tensile or burst strength), or another quantitative degradation-related endpoint.

Term	Meaning in this app and guide
Scaling	Preprocessing that changes the numerical scale of time or response values, such as applying a base-10 logarithm, without changing which observations are present.
Scientific plausibility	Consistency of the fitted relationship, parameter values, and predicted behavior with relevant knowledge of the material, degradation mechanism, test conditions, and intended use.
Shared-grid interpolation	Interpolation option that estimates response values at the union of observed time points and estimates times at the union of observed response values. Estimates outside the overlapping range are unavailable.
Standard deviation (SD)	Spread of individual measurements at a given time. For Approach 2, supply SD when each entered response is an individual observation; use SEM when each response is a reported mean.
Standard error (SE; parameter)	Approximate standard deviation of a fitted parameter estimate implied by its covariance. It describes parameter precision and differs from SEM, which describes uncertainty in a reported response mean.
Standard error of the mean (SEM)	Uncertainty in a reported mean, expressed as the standard deviation of that mean. Approach 2 uses supplied SEM directly; an optional replicate count does not divide it again.
Test data	Observations set aside during a CV split and used to evaluate predictions from a model fitted without those observations.
Time constant (integral time constant)	Characteristic time scale calculated, when finite, as the area under a fitted decay curve after normalizing its fitted value at time zero to 1, $\tau = f(0)^{-1} \int_0^\infty f(t)\,dt$. In Approach 2, in vitro and in vivo time constants can be compared or used as a ratio for time rescaling.
Training data	Observations retained during a CV split and used to fit the model for that split.
Weight (information criterion)	Normalized relative-support value calculated from information-criterion deltas across the displayed candidate set. Weights sum to 1 within that set and are not absolute probabilities that a model is correct.

References

[1] U.S. Food and Drug Administration. Assessing the credibility of computational modeling and simulation in medical device submissions: Guidance for industry and food and drug administration staff. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-device-submissions, November 2023.

[2] U.S. Food and Drug Administration. Reporting of computational modeling studies in medical device submissions: Guidance for industry and food and drug administration staff. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/reporting-computational-modeling-studies-medical-device-submissions, September 2016.

[3] William T. Coffey and Yuri P. Kalmykov. *The Langevin Equation: With Applications to Stochastic Problems in Physics, Chemistry and Electrical Engineering*. World Scientific, 4 edition, 2017.

[4] Harry B. Hopfenberg. Controlled release from erodible slabs, cylinders, and spheres. In *Controlled Release Polymeric Formulations*, volume 33 of *ACS Symposium Series*, pages 26–32. American Chemical Society, 1976. https://doi.org/10.1021/bk-1976-0033.ch003.

[5] A. Tsoularis and J. Wallace. Analysis of logistic growth models. *Mathematical Biosciences*, 179(1):21–55, 2002. https://doi.org/10.1016/S0025-5564(02)00096-2.

[6] M. H. Zwietering, I. Jongenburger, F. M. Rombouts, and K. van 't Riet. Modeling of the bacterial growth curve. *Applied and Environmental Microbiology*, 56(6):1875–1881, 1990. https://doi.org/10.1128/aem.56.6.1875-1881.1990.

[7] Toma-Leonida Dragomir, Ana-Maria Pană, Valentin Ordodi, Vasile Gherman, Gabriela-Alina Dumitrel, and Sorin Nanu. An empirical model for predicting biodegradation profiles of glycopolymers. *Polymers*, 13(11):1819, 2021. https://doi.org/10.3390/polym13111819.

[8] Achim Göpferich. Polymer bulk erosion. *Macromolecules*, 30(9):2598–2604, 1997. https://doi.org/10.1021/ma961627y.

[9] Susanne Fredenberg, Marie Wahlgren, Mats Reslow, and Anders Axelsson. The mechanisms of drug release in poly(lactic-co-glycolic acid)-based drug delivery systems–a review. *International Journal of Pharmaceutics*, 415(1–2):34–52, 2011. https://doi.org/10.1016/j.ijpharm.2011.05.049.

[10] S. Dean Allison. Analysis of initial burst in PLGA microparticles. *Expert Opinion on Drug Delivery*, 5(6):615–628, 2008. https://doi.org/10.1517/17425247.5.6.615.

[11] Timothy O. Hodson. Root-mean-square error (RMSE) or mean absolute error (MAE): When to use them or not. *Geoscientific Model Development*, 15:5481–5487, 2022. https://doi.org/10.5194/gmd-15-5481-2022.

[12] Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. https://doi.org/10.1016/j.ijforecast.2006.03.001.

[13] Jeremy Miles. R squared, adjusted r squared. In *Wiley StatsRef: Statistics Reference Online*. John Wiley & Sons, 2014. https://doi.org/10.1002/9781118445112.stat06627.

[14] Sylvain Arlot and Alain Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010. https://doi.org/10.1214/09-SS054.

[15] Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012. https://doi.org/10.1016/j.ins.2011.12.028.

[16] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. https://doi.org/10.1111/ecog.02881.

[17] Christoph Bergmeir, Rob J. Hyndman, and Bonsoo Koo. A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Computational Statistics & Data Analysis*, 120:70–83, 2018. https://doi.org/10.1016/j.csda.2017.11.003.

[18] NIST/SEMATECH. Weighted least squares. e-Handbook of Statistical Methods, Section 4.4.3.2. https://www.itl.nist.gov/div898/handbook/pmd/section4/pmd432.htm.

[19] The SciPy community. `scipy.optimize.curve_fit`. SciPy Reference Guide. https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.curve_fit.html.

[20] Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. https://doi.org/10.1109/TAC.1974.1100705.

[21] Kenneth P. Burnham and David R. Anderson. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. Springer, 2nd edition, 2002. https://doi.org/10.1007/b97636.

[22] Eric-Jan Wagenmakers and Simon Farrell. AIC model selection using akaike weights. *Psychonomic Bulletin & Review*, 11:192–196, 2004. https://doi.org/10.3758/BF03206482.

[23] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. https://doi.org/10.1214/aos/1176344136.

[24] Clifford M. Hurvich and Chih-Ling Tsai. Regression and time series model selection in small samples. *Biometrika*, 76(2):297–307, 1989. https://doi.org/10.1093/biomet/76.2.297.

[25] Robert S. Maier. Information criteria for deciding between normal regression models, 2013. Preprint, version 2. https://arxiv.org/abs/1305.5493v2.

[26] Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995. https://doi.org/10.1080/01621459.1995.10476572.

[27] Michael H. Kutner, Christopher J. Nachtsheim, John Neter, and William Li. *Applied Linear Statistical Models*. McGraw-Hill/Irwin, 5 edition, 2005.

[28] Harvey J. Motulsky and Lennart A. Ransnas. Fitting curves to data using nonlinear regression: A practical and nonmathematical review. *The FASEB Journal*, 1(5):365–374, 1987.

[29] Douglas M. Bates and Donald G. Watts. *Nonlinear Regression Analysis and Its Applications*. Wiley Series in Probability and Statistics. John Wiley & Sons, 1 edition, 2007.

[30] Georg Heinze, Christine Wallisch, and Daniela Dunkler. Variable selection – a review and recommendations for the practicing statistician. *Biometrical Journal*, 60(3):431–449, 2018. https://doi.org/10.1002/bimj.201700067.

[31] NIST/SEMATECH. Graphical residual analysis—initial model. NIST/SEMATECH e-Handbook of Statistical Methods, 2022. https://www.itl.nist.gov/div898/handbook/pmd/section6/pmd614.htm.

[32] Penn State Eberly College of Science. Further residual plot examples. STAT 462: Applied Regression Analysis. https://online.stat.psu.edu/stat462/node/124/.

[33] OriginLab Corporation. Residual plot analysis. OriginLab Documentation. https://docs.originlab.com/origin-help/residual-plot-analysis/.